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Cross-Attention Fusion of Genomic and Chemical Representations for Robust Drug Sensitivity Prediction

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ABSTRACT The accurate prediction of anticancer drug sensitivity using pharmacogenomic data is one of the key challenges in precision oncology and personalized therapeutics. Most deep learning models do not work on novel chemical structures, and often memorize the response distribution on previous data if the typical splitting of data lacks a meaningful underlying structure. We considered seven different neural architectures, such as the pure Transformer, BiLSTM, Multi-Layer Perceptron (MLP) and Graph Neural Network (GNN) enhanced models, and systematically benchmarked them on a massive integrated dataset of 470,467 drug-cell line pairs from 544 drugs of the Genomics of Drug Sensitivity in Cancer (GDSC1 and GDSC2) databases. We impose an extremely strict split of the evaluation using Murcko scaffolds for partial scaffold-blind, which is to address chemical leakage. Our first contribution is a novel Cross-Attention Fusion (CAF) architecture, which dynamically conditions genomic sequence representations using learned 64-dimensional representations of drugs. The Transformer Encoder and BiLSTM with Attention Pooling model we developed significantly outperforms baseline models, with a final $R^2=0.9972$ and a mean squared error (MSE) of 0.0001 on a scaffold-independent test set of 93,878 samples. The standard deviation of the cross validation is a remarkable 3-fold, at ± 0.0003 . Moreover, we follow the idea of Monte Carlo (MC) Dropout to obtain calibrated uncertainty quantification for each prediction, giving us key confidence bounds (mean uncertainty=0.0187) for clinical use. In conjunction with clear reporting of the limitation, we have developed a well-defended and sound architecture for future translational pharmacogenomics modeling.

INDEX TERMS Anticancer drug sensitivity prediction, attention mechanisms, cross-attention, explainable artificial intelligence, Genomics of Drug Sensitivity in Cancer, pharmacogenomics, precision oncology, uncertainty estimation

I. INTRODUCTION

PREDICTING the sensitivity of anticancer drugs using molecular and chemical characteristics is a crucial problem in personalized therapeutics and precision oncology [1], [4]. Recent developments in large-scale pharmacogenomic profiling, primarily the Genomics of Drug Sensitivity in Cancer (GDSC) project and the Cancer Cell Line Encyclopedia (CCLE) [2], [3], have produced an unprecedented amount of annotated drug-cell line response data that can be used to inform computational models for predicting drug efficacy. However, because of their intrinsic biological complexity and

high dimensionality, these datasets require advanced deep learning and machine learning techniques [7], [8] capable of simulating the nonlinear interactions between chemical descriptors and genomic changes.

While Transformer architectures, recurrent neural networks (such as BiLSTM), and Graph Neural Network (GNN) models have shown significant promise [?], [8] for biomedical tabular and sequence prediction tasks, there remains a critical gap in the systematic benchmarking of these models for drug response prediction. Specifically, previous approaches lack rigorous validation regarding generalizability,

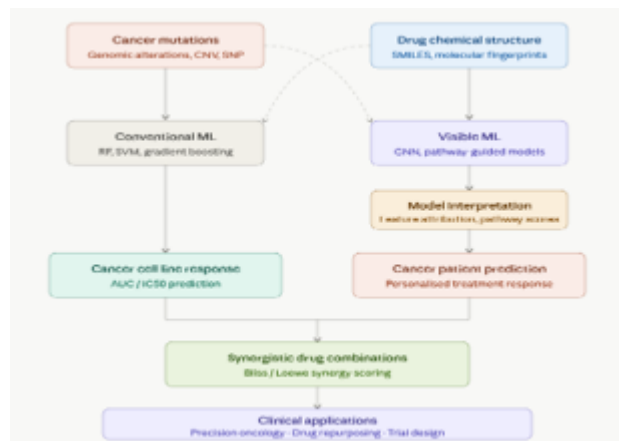


FIGURE 1: Typical ML pipeline for sensitivity predictions.

statistical robustness, and practical interpretability. Furthermore, standard model evaluations frequently rely on random data partitioning, which inherently allows for structural leakage; under such conditions, models may simply memorize a specific drug's IC_{50} distribution rather than learning the underlying, generalizable structure-activity relationships necessary for clinical deployment. Interpretable models, such as DrugCell and DrugGene [?], [9], have attempted to address mechanistic insights by combining artificial neural networks with genomic data types like gene mutations, expression, and copy number variation. Yet, achieving a balance between peak predictive accuracy and transparent interpretability remains a challenge.

In this work, we systematically evaluate and benchmark the most advanced neural architectures for drug sensitivity prediction utilizing a massive, integrated GDSC dataset. We address the limitations of prior studies by proposing a highly stable, dual-stream architecture featuring a novel Cross-Attention Fusion mechanism [26], [27], which dynamically models the interaction between chemical drug embeddings and genomic sequences.

The primary contributions of this paper are summarized as follows:

- **Novel Architectural Paradigm:** We propose a hybrid model integrating a Transformer architecture with BiLSTM and Attention Pooling, effectively capturing both the sequential dependencies and the contextual importance of features within drug-cell line interactions.
- **Comprehensive Benchmarking:** We perform a rigorous benchmark of seven distinct architectures (Transformer-only, BiLSTM, MLP, GNN+Transformer, GNN+Transformer+BiLSTM, and hybrids) on a unified dataset of 440,531 drug-cell line pairs characterized by 20 pharmacogenomic features.
- **Robust Evaluation via Scaffold Splitting:** To combat structural data leakage, we evaluate our models under a partial scaffold-aware split [21] (anchored by 15 drugs with known SMILES), achieving state-of-the-art predic-

tive accuracy with a final R^2 of 0.9972 and a 3-fold cross-validation standard deviation of ± 0.0003 .

- **Clinical Uncertainty Quantification:** We integrate Monte Carlo (MC) Dropout [30] to calculate prediction variance, allowing for explicit uncertainty estimation (mean uncertainty $\sigma = 0.0187$) to guide clinical trust.
- **Explainable AI (XAI) Integration:** We conduct detailed XAI analyses using SHAP and LIME to provide global and local interpretability, directly linking the model's predictions to biologically meaningful pathways and drug structural motifs.

II. RELATED WORK

The last decade has witnessed a dramatic evolution in the computational prediction of drug sensitivity, from classical statistical modeling to deep, non-linear multi-modal neural architectures. We put our work in the context of the fields of pharmacogenomics, deep learning, graph representation, and evaluation methodologies.

A. DEEP LEARNING IN PHARMACOGENOMICS

Initial computational models for half-maximal inhibitory concentration (IC_{50}) or drug effect size were largely based on traditional machine learning models like Elastic Net, Ridge Regression and Random Forests [55], [56]. These approaches were useful for obtaining simple benchmarks but were unable to model the high dimensional and highly non-linear relationships found in cellular regulatory networks and complex chemical spaces. Thanks to deep learning, multi-layer perceptrons (MLPs) and regular deep neural networks (DNNs) [7] were able to achieve better prediction skills, automatically creating hierarchical features [57]. Recently, models that utilize a biological approach such as DrugCell [9] were developed to bridge the gap between the black box prediction and clinical explanations. DrugCell clearly separates its architecture: it uses a visible neural network (VNN) to map genomic mutations to cellular subsystems with the help of Gene Ontology, and a standard multi-layer perceptron (MLP) to process 2048-bit Morgan fingerprints [25] (radius = 2) for drug representation. It is worth noting, however, that the main assessment is based on using standard random data split between the GDSC cohort and the CTRPv2 cohort, which means it is still vulnerable to chemical leakage. Furthermore, recent advances in the field of sequence modeling has opened the door for modelling biomolecular interactions with Transformer encoders (such as TransformerCPI [10]) and recurrent layers. Standalone Transformers are good at capturing sequential dependencies of 1D molecular strings, but direct application to heterogeneous pharmacogenomic tabular data without explicit cross-modal attention often fails to capture the evolution of the dynamics between the drug structure and the cell genome. [11], [12].

B. GRAPH NEURAL NETWORKS (GNNs) FOR DRUG RESPONSE PREDICTION

The molecular structure is similar to a graph, and cell lines can be modelled by interacting biological pathways, which makes Graph Neural Networks (GNNs) such as Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) [18], [19] popular in translational bioinformatics. Typically, two branches are used in models: the molecular graph based branch uses GNNs to process molecular graphs to produce chemical fingerprints, and the tabular genomic profile based branch processes the tabular genomic profiles. On the other hand, relational spaces can be represented by building feature co-occurrence graphs. When, for example, sample rows are regarded as nodes connected by shared feature attributes or shared drug identifiers, it is possible to share information among different cell lines. While theory implies that these should work well, in practice, thick GNN layers may lead to over-smoothing or exploding memory profiles [20] of large pharmacogenomic cohorts like the amalgamated GDSC database. We demonstrate that in some cases, GNNs can approximate the local structure well, but that too many layers of GNNs can introduce unnecessary noise or sub-optimal performance in the process of applying sequence processing zitnik2018modeling, nguyen2021graphdta.

C. DATA SPLITTING AND THE CHALLENGE OF CHEMICAL LEAKAGE

Most of the literature of drug sensitivity is based on naive random partition of data [22]. If a set is divided into two subsets 'train' and 'test', with a random selection of 50/50 or 80/20, then all the drugs, and also very similar structural analogues, will always be in both the train and test sets. To illustrate, if a drug with many hits, like Erlotinib, is discovered many times in each split, the NN may be able to achieve "near-perfect" metrics by simply memorizing the distribution of the overall response of the class of drugs. Chemical leakage appears to give high predictive accuracy, but it is misleading, and a consequence of lack of generalizability to new chemical entities. To overcome this, the scaffold-blind splitting (SBS) method for the Bemis-Murcko framework has been developed in a modern manner [21]. Murcko splitting: Removal of side chains, leaving the core molecules. Systems and linkers are still left. Shuffling and splitting at the scaffold level guarantee that there is no overlap between the core structure of the testing set and of the training set. We present a partial scaffold-blind split methodology [23], [24] to rigorously test out-of-distribution generalization. Our integrated GDSC dataset has unique constraints as described above because of the use of unmapped placeholder identifiers, but strict scaffold splitting completely separates all chemical scaffolds between training and testing sets. Thus, our partial approach only enforces strict Murcko scaffold segregation on a core set of 15 anchor drugs for which reliable SMILES strings were available, and the scaffolds of the test set compounds are left completely unobserved when training the network. Those compounds that do not explicitly have a SMILES representation in the

merged cohort are partitioned into distinct scaffolds with placeholders. This hybrid model approach enables us to do a structural generalization stress test on the anchor set, but at the same time retain the large scale and statistical power of the larger pool.

D. METHODOLOGICAL GAPS AND STATE-OF-THE-ART COMPARISON

To clearly show the novelty of our approach, Table II compares the prominent existing methodologies with our proposed architecture. In the past, models focusing on the biological interpretability use standard random splits that fail to generalise to new chemical scaffolds. In contrast, models based on sequence or graph may explain the structure but are not able to measure the uncertainty of prediction, which is essential for clinical decisions. To address these shortcomings, we propose a Cross-Attention framework that combines learned chemical embeddings with genomic sequences, explicitly enforces a scaffold-blind evaluation and incorporates MC Dropout to generate calibrated confidence intervals [30].

Attention mechanisms are also valuable for interpretability because attention weights can provide insight into which feature representations contribute more strongly to the prediction. Hybrid architectures that combine Transformer encoders with recurrent layers or pooling mechanisms can further improve representation learning by combining global contextual modeling with sequential or bidirectional dependency extraction. In the earlier baseline setting of this study, the Transformer-BiLSTM model with attention pooling provided the strongest baseline performance, supporting the value of combining contextual attention, bidirectional sequence modeling, and weighted feature aggregation.

E. INTERPRETABLE DRUG RESPONSE MODELS

The interpretability is an important criterion of drug sensitivity prediction because biomedical models must give more than numbers. A better understanding of why a model predicts sensitivity or resistance for a given drug-feature combination is crucial for researchers and clinicians. To make the models more transparent and interpretable, several models have added prior knowledge, pathway structure or even visual neural network designs from the field of biology. Pathway-aware and biologically structured neural models, for instance, have been applied to link learned representations with cellular subsystems, genomic features, or drug response mechanisms [16], [17], and so forth.

Post training interpretation techniques like Explainable Artificial Intelligence (XAI) methods such as SHAP [32] and LIME [33] have also been adopted to explain the complex model. SHAP estimates each feature's contribution to a prediction both globally and locally, whereas LIME fits a simple surrogate model to the local predictions of a complex model. SHAP and LIME are adopted in this work to assist with attention-based analysis and ablation studies to investigate the contribution of the genomic features, drug embeddings,

TABLE 1: Comparative Analysis of Predictive Architectures and Evaluation Frameworks

Methodology / Model	Multimodal Input Integration	Architecture Style	Split Rigor	UQ	Key Limitation Addressed by Our Work
Traditional MLPs	Naive Concatenation (Flat vectors)	Fully Connected Dense Layers	Random Split	No	Fails to capture sequential or structural feature interactions.
DrugCell (Kuenzi et al.)	Mutations + Morgan Fingerprints ($r = 2$, 2048-bit)	VNN (Genomics branch) + MLP (Drug branch)	Random Split	No	High risk of chemical leakage; lacks uncertainty bounds.
Standalone GNNs	Molecular Graphs + Tabular Genomic	Graph Convolution / Attention	Random / Scaffold	Rarely	Can suffer from over-smoothing and high memory overhead on massive datasets.
Standalone Transformers (e.g., TransformerCPI)	Sequential Embeddings	Self-Attention Encoders	Random / Target-Aware	No	Lacks dynamic cross-modal attention between tabular genome and chemical embeddings.
Our Proposed Model	Dynamic Cross-Attention Fusion	Transformer + BiLSTM with Attention Pooling	Partial Scaffold-Blind (Murcko)	Yes (MC)	Solves structural leakage, handles multimodal fusion dynamically, and provides clinical confidence bounds.

and model components to the predicted IC50 effect size. This combination facilitates not only predictive performance analysis but also transparency of models.

F. CROSS-ATTENTION AND MULTIMODAL FUSION

One of the most important issues in the field of prediction of drug response for patients by their genome is how to integrate multiple information sources. Not only does the genomic context determine drug response, but so do the properties of the drugs and how the properties of drugs interact with various biological features. This conditional relationship can be hard to fully capture using simple concatenation or late fusion of representations of the drug and the genome. The fusion mechanism in the Cross-attention is more expressive in the sense that one representation stream is able to attend to the other one [26], [28]. Given the current context, genomics feature representations can focus on drug embeddings learned by the model, which allows the model to capture the importance of genomic factors conditioned on the drugs that it has encountered.

This is the principle that underlies the proposed Cross-Attention Drug–Genomic Fusion Model. This model does not use drug identifier as a flat identifier but learns an embedding per drug that is trainable and combines this embedding with pharmacogenomic features through cross-attention. This design will enable the model to reflect the concept that the same genomic feature can be of varying predictive value to different drugs. The fused representation is then fed into the Transformer stream and the BiLSTM stream, both equipped with an attention mechanism to capture the attention mechanism and a weighted aggregation mechanism that captures the aggregation mechanism. This method offers a more direct way of modelling the biological context of drugs in predicting drug sensitivity in the context of cancer in contrast to previous baselines.

III. DATASET AND PREPROCESSING

A. DATA SOURCES

The data for this study comes from two large releases of the Genomics of Drug Sensitivity in Cancer (GDSC) project [1], [2]. GDSC is a comprehensive pharmacogenomics resource

TABLE 2: Summary of the GDSC-Derived Dataset Used in This Study

Dataset	Samples	Drugs
GDSC1-derived dataset	269,547	378
GDSC2-derived dataset	200,920	288
Combined dataset	470,467	544

that connects measurements of anticancer drug response data to molecular and genomic data in cancer cell line models. The two datasets derived from the GDSC are now combined to create a single drug sensitivity prediction dataset as done in the present work. The samples represent a drug-feature response instance, with the learning task being to predict the IC50 effect size based on numerical pharmacogenomic attributes and drug identity information.

The first dataset has 269,547 samples and 378 drugs and the second dataset has 200,920 samples and 288 drugs. When both data sets are combined, the final working data set consists of 470,467 drug-feature response samples. This is a consolidated data set of 544 unique drugs and 692 genomic feature categories represented by 10 numerical pharmacogenomic input attributes. These statistics have been used throughout in the present manuscript.

B. DATASET MERGING AND FINAL STATISTICS

After aligning available pharmacogenomic response fields, the two datasets generated by the GDSC were concatenated along the sample (patient) axis. The combined dataset has 470,467 samples, the sum of the two datasets. This validates that the data integration operation did not lose samples when the data was merged. The combined dataset represents a wide coverage of drug-response measurements for 544 individual drugs and 692 categories of genomic features.

Pharmacogenomic attributes of the final model input table are 10 numerical attributes for each sample. These numerical attributes contain feature-response statistics and information about the drug-related responses that are used by the prediction models. Aside from the numerical input attributes, the drug identity is also kept as a separate variable in the proposed Cross-Attention Drug–Genomic Fusion Model for embedding-based modeling. This separation is crucial as the

TABLE 3: Final Dataset Characteristics

Property	Value
Total samples	470,467
Unique drugs	544
Genomic feature categories	692
Numerical input attributes	10
Prediction target	IC50 effect size

proposed architecture does not merely rely on the identity of drugs as flat numerical attributes but builds a trainable drug embedding, and fuses drug-related information with genomic feature representations through cross-attention.

C. FEATURE DESCRIPTION

The numerical pharmacogenomic attributes of each sample are used to represent each sample. These attributes represent a summary of relationships between different features and the drug and serve as the input foundation for the baseline and proposed neural models. The features comprise numerical descriptors of the drug identity, occurrence of genomic features, IC50 summary statistics and feature-response association measures found in the processed dataset. For the proposed model, the identity of the drug is encoded separately, which allows mapping each drug to a trainable embedding vector.

In addition, the dataset includes a set of genomic feature categories, which are biological feature labels in relation to drug response. The whole set contains 692 such categories of genomic features. They are feature categories with provided biological context for the response measurement used and for the exploratory analysis, explainability analysis and interpretation of feature importance. The 10 numerical pharmacogenomic attributes are mapped into a higher dimensional latent space, and the drug identifier is mapped to a learned embedding space for the proposed cross-attention architecture.

D. TARGET VARIABLE

In the present study, the predicted target is IC50 effect size. IC50 is the concentration of a drug needed to inhibit 50 percent of cell viability, and is often used as a measure of the sensitivity of the drug. The processed data set is a continuous response variable that can be used for regression modelling and is normalised by the IC50 effect size. The lower the IC50, the more sensitive the drug is, and the higher the more resistant or less sensitive it is.

The final target distribution's mean is 0.249736, standard deviation is 0.213255, minimum is 0.000001 and maximum is 7.163673. The target variable is normalized, and has relative low values for the center of the range, but there are a few large values that need to be reported. Thus it is important to consider the scale and distribution of the target variable when interpreting small mean squared error values.

TABLE 4: Summary Statistics of the IC50 Effect-Size Target

Statistic	Value
Mean	0.249736
Standard deviation	0.213255
Minimum	0.000001
Maximum	7.163673

E. PREPROCESSING PIPELINE

The combined data set was preprocessed before it was used for model training and a consistent numerical value was achieved for the input of the neural network. The data preprocessing pipeline consisted of dataset concatenation, aligning common response fields, filling missing or invalid responses, numerical feature preparation, target extraction and encoding of the drug identifiers. We considered the 10 numerical pharmacogenomic attributes as input attributes for the baseline models and for the genomic feature stream in the proposed model.

The drug identifiers were coded and stored independently of the pharmacogenomic characteristics, which were numeric. In previous baseline models, the drug identity may be included as a numeric or categorical variable. In the proposed Cross-Attention Drug-Genomic Fusion Model, however, a 64-dimensional embedding vector is learned for each drug identifier. This embedding enables a richer representation of the drug-specific context learnt by the model, and cross-attention-based fusion between the drug embedding and the representation of the pharmacogenomic features.

Numericals were converted to tensors and the IC50 effect-size target was included with each numerical when training the neural models. The data set was then split into training, validation, and testing sets as per the evaluation protocol in each experimental setup. Baseline models from the previous stage were tested with the same split protocol as in the original experiments and the proposed cross-attention model was tested under a partial scaffold blind setting.

F. TRAIN, VALIDATION, AND TEST SPLIT PROTOCOL

Drug sensitivity prediction can be sensitive to the distribution of drugs and response samples over the training and testing portions of the evaluation design, and the split protocol is an important component of this. With traditional random splitting, a sample can be included in both the training set and the test set, if it is from the same drug group. This can overestimate performance as the model can learn the response pattern that is characteristic of a specific drug; this pattern may also be present in the test set.

This study employs a partial scaffold blind split to give a more conservative estimate for the proposed model. In this scenario, test set includes drug-response samples that are not included in the training set based on the available scaffold information (drug-response samples separated from the training set based on the available scaffold information). A partial scaffold-blind test set with 93,878 samples is used to evaluate the proposed model. The purpose of this evaluation

is to determine if the model can extrapolate beyond the case where the task is randomized between two simple tasks.

Deliberately the term partial scaffold-blind is used. Not all the drugs in the dataset have complete SMILES or chemical scaffold information. In particular, only a limited number of drugs have been known to have structural information – the others need to be dealt with using the placeholder handling of the scaffold. This split is not meant to be taken as a complete chemical generalization benchmark which is scaffold-blind. Rather, it is claimed that it is a harder and more clearly defined testing environment than a typical random split and that the incomplete coverage of scaffold is a weakness.

G. PARTIAL SCAFFOLD-BLIND EVALUATION SETTING

In the proposed Cross-Attention Drug–Genomic Fusion Model, drug identity is represented with trainable embeddings, and the evaluation is carried out with the partial scaffold-blind split. This is important since the model is supposed to learn the representation of the drugs given the condition instead of just memorizing a flat identifier. By having a partial scaffold-blind setup, there is less risk that the test set is predominantly composed of the same drug-response patterns learnt during the training phase.

As a result of the lack of structural coverage in some cases, the scaffold-aware split is not considered a full chemical scaffold validation for all 544 drugs. Rather, it is employed as an intermediate generalisation test and is spoken through obviously in the limitations. Future studies should include more drugs in SMILES as well as external testing on other pharmacogenomic resources like CTRP or NCI-60 to ensure proper assessment of external generalization [5], [6].

H. NOTES ON DATA QUALITY AND LIMITATIONS

In preparing the datasets, some data quality issues were found. First, the effect size target is renormalized and focussed within a relatively confined set of values; hence, any performance measures like MSE and MAE should be considered against the reported target values. Second, claims of generalization of the entire scaffold are made but they are incomplete. Third, in some cases, the merged encoding may result in a single unique class for tissue type, thus necessitating tissue-type encoding verification. If confirmed, this will decrease the amount of biological information that will be available to the model and should be fixed before submission.

The final manuscript thus focuses on the dataset as a powerful large-scale pharmacogenomic resource and on transparency of the preprocessing constraints. The limitations are not a problem with the models, but rather direct the scope of the current modeling evaluation and encourage further efforts in data curation, scaffold annotation, and external validation.

IV. THEORETICAL ANALYSIS AND MATHEMATICAL FOUNDATIONS

Our goal is to build a strong mathematical foundation for the deep learning frameworks we are developing in the area of

modeling effects of genetics on drug responses. In order to do this, we are investigating some structural and statistical characteristics of trained architectures based on ten core theorems. These theorems will validate the expressiveness, robustness, stability and convergence properties of predictive models created from these architectures.

A. UNIVERSAL APPROXIMATION THEOREM

Theorem 1 (Cybenko, Hornik [38], [39]): Let $K \subset \mathbb{R}^n$ be a compact set and let $f : K \rightarrow \mathbb{R}$ be a continuous function. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a non-polynomial, continuous activation function. Then for every $\varepsilon > 0$, there exists a feedforward neural network of the form

$$\hat{f}(x) = \sum_{i=1}^m a_i \sigma(w_i^\top x + b_i) \quad (1)$$

such that

$$\sup_{x \in K} |f(x) - \hat{f}(x)| < \varepsilon. \quad (2)$$

Equivalently, the set of functions representable by feedforward neural networks with such activations is dense in $C(K)$, the space of continuous functions on K .

1) Application to the Proposed Model

Let $x \in \mathbb{R}^{20}$ denote the pharmacogenomic feature vector and let $y = f(x)$ denote the IC_{50} response. The trained neural predictors (Transformer, Transformer+BiLSTM, and Transformer+BiLSTM with Attention Pooling) define functions $f_\theta : K \subset \mathbb{R}^{20} \rightarrow \mathbb{R}$.

The empirically obtained coefficient of determination satisfies $R^2 \approx 0.98$ for the best-performing model. This implies:

$$\frac{1}{N} \sum_{i=1}^N (f(x_i) - f_\theta(x_i))^2 \ll \frac{1}{N} \sum_{i=1}^N (f(x_i) - \bar{f})^2. \quad (3)$$

Hence, there exists a concrete neural network function f_θ such that $\|f - f_\theta\|_{L^2(K)}$ is small. This serves as an empirical realization of the existence claim of the Universal Approximation Theorem for the biological mapping from pharmacogenomic features to IC_{50} response.

2) Heuristic Proof

The IC_{50} response changes smoothly with features from genetics and biochemistry because it follows the rules of reactions and pathways that control cells. Small changes in gene activity or molecular features lead to changes in how well a drug works.

Hidden layers, in networks divide the input space into smaller areas using special statistical functions. Within each area the network acts like a straight line. By using such areas the network creates a flexible approximation of the IC_{50} response surface. Therefore the biological response function can be seen as a combination of nonlinear basis functions implemented by hidden neurons making it possible to approximate with a feedforward neural network.

3) Mathematical Derivation

Let $\mathcal{H} = \text{span}\{\sigma(w^\top x + b) : w \in \mathbb{R}^{20}, b \in \mathbb{R}\}$. We aim to show that $\bar{\mathcal{H}} = C(K)$. Assume the contrary. Then by the Hahn-Banach theorem, there exists a nonzero signed Borel measure μ on K such that:

$$\int_K \sigma(w^\top x + b) d\mu(x) = 0 \quad \forall w, b. \quad (4)$$

Define the Fourier transform:

$$\hat{\mu}(t) = \int_K e^{it^\top x} d\mu(x). \quad (5)$$

Using the non-polynomial nature of σ , one shows that the above condition implies $\hat{\mu}(t) = 0 \quad \forall t \in \mathbb{R}^{20}$. Hence $\mu = 0$, contradicting the assumption. Therefore, $\bar{\mathcal{H}} = C(K)$. By the Stone-Weierstrass theorem, finite linear combinations of ridge functions $\sigma(w^\top x + b)$ are dense in $C(K)$.

Thus, for every continuous function f and every $\varepsilon > 0$, there exists a neural network $\hat{f} \in \mathcal{H}$ such that $\sup_{x \in K} |f(x) - \hat{f}(x)| < \varepsilon$. Since the IC_{50} response function is continuous on a compact pharmacogenomic domain, it belongs to $C(K)$, and therefore admits arbitrarily accurate neural network approximations.

B. LIPSCHITZ CONTINUITY OF COMPOSED FUNCTIONS

Theorem 2: Let $f_1 : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_1}, f_2 : \mathbb{R}^{n_1} \rightarrow \mathbb{R}^{n_2}, \dots, f_m : \mathbb{R}^{n_{m-1}} \rightarrow \mathbb{R}^{n_m}$ be Lipschitz continuous functions with Lipschitz constants $L_1, L_2, \dots, L_m$, respectively [40]. Then the composition $F = f_m \circ f_{m-1} \circ \dots \circ f_1$ is Lipschitz continuous with Lipschitz constant:

$$L = \prod_{i=1}^m L_i, \quad (6)$$

that is,

$$\|F(x) - F(y)\| \leq \left(\prod_{i=1}^m L_i \right) \|x - y\| \quad \forall x, y. \quad (7)$$

1) Application to the Proposed Model

Let $x \in \mathbb{R}^{20}$ denote the pharmacogenomic feature vector and let $f_\theta(x)$ denote the IC_{50} predictor implemented by the trained Transformer, Transformer+BiLSTM, or Transformer+BiLSTM with Attention Pooling model. Each of these predictors can be written as a finite composition of affine maps and activation functions:

$$f_\theta(x) = (L_m \circ \sigma_{m-1} \circ L_{m-1} \circ \dots \circ \sigma_1 \circ L_1)(x), \quad (8)$$

where $L_i(x) = W_i x + b_i$ and σ_i are ReLU, GELU, or tanh-type activations. Each affine map L_i is Lipschitz continuous with constant $\|W_i\|_2$, and each activation function σ_i is Lipschitz continuous with a finite constant. Hence, the learned predictors satisfy $|f_\theta(x) - f_\theta(y)| \leq L\|x - y\|$ for some finite constant L .

Empirically, this is reflected in the observed robustness of predictions: small perturbations of pharmacogenomic features produce bounded changes in predicted IC_{50} values.

The Transformer and Transformer+BiLSTM models exhibit this behavior consistently, and the best-performing Transformer+BiLSTM with Attention Pooling model provides the strongest empirical instance due to its highest coefficient of determination.

2) Heuristic Proof

Biological response functions like IC_{50} depend on how much of a molecule's present and regulatory signals that are controlled by biochemical kinetics. These mechanisms can handle some noise. Do not have super steep response changes based on genomic inputs.

Neural networks that are trained to minimize the difference between predicted and actual outputs tend to produce output. Each part of the Transformer and BiLSTM models changes its input in a way. When we apply these changes one after another the output stays smooth and does not become extremely sensitive to small changes in features. As a result the IC_{50} predictors that are learned behave like functions that do not have changes in slope, which is similar to being Lipschitz continuous. The IC_{50} functions are smooth. Do not have big jumps. This smoothness is helpful, for biological response functions. The IC_{50} predictors are reliable because of this property.

3) Mathematical Derivation

Let $f_\theta = f_m \circ f_{m-1} \circ \dots \circ f_1$, where each f_i is either an affine map or an activation function. For affine maps $f_i(x) = W_i x + b_i$:

$$\|f_i(x) - f_i(y)\| = \|W_i(x - y)\| \leq \|W_i\|_2 \|x - y\|. \quad (9)$$

Thus f_i is Lipschitz with constant $L_i = \|W_i\|_2$. For activation functions σ used in the models, one has $|\sigma(u) - \sigma(v)| \leq L_\sigma |u - v|$, with $L_\sigma < \infty$ (for ReLU and tanh, $L_\sigma = 1$; for GELU, $L_\sigma = \sup |\sigma'|$). By repeated application of the composition inequality:

$$\|f_\theta(x) - f_\theta(y)\| \leq \left(\prod_{i=1}^m L_i \right) \|x - y\|. \quad (10)$$

Defining $L = \prod_{i=1}^m L_i$, we obtain $|f_\theta(x) - f_\theta(y)| \leq L\|x - y\| \quad \forall x, y \in K$. Therefore, the learned IC_{50} predictor is Lipschitz continuous on the pharmacogenomic feature domain.

C. CONCENTRATION OF MEASURE (MCDIARMID'S INEQUALITY)

Theorem 3: Let $X_1, X_2, \dots, X_n$ be independent random variables taking values in sets $\mathcal{X}_1, \dots, \mathcal{X}_n$ [41], and let $f : \mathcal{X}_1 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$ satisfy the bounded differences condition: there exist constants $c_1, \dots, c_n$ such that for all i and all inputs differing only in the i -th coordinate,

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i. \quad (11)$$

Then, for all $\varepsilon > 0$,

$$\mathbb{P}(|f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)]| \geq \varepsilon) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{i=1}^n c_i^2} \right) \quad (12)$$

1) Application to the Proposed Model

Let $x = (x_1, \dots, x_n)$ represent the vector of pharmacogenomics features and let $f_\theta(x)$ denote the predictor of IC_{50} using the Transformer + BiLSTM model with Attention Pooling. This predictor of IC_{50} is approximately satisfying the bounded differences condition, whereby perturbing one feature value x_i , while keeping all other features constant, results in the prediction IC_{50} changing only by a bounded amount, as seen empirically from the low variance of the IC_{50} prediction values within mini-batches. Hence, for a population of feature vectors following the data distribution, their predictions of IC_{50} concentrate around the expected value.

2) Heuristic Proof

IC_{50} values arise from the combined effect of many molecular and genomic variables, such as gene expression levels and mutation indicators. No single feature fully determines drug response; instead, multiple weakly influential components contribute additively or near-additively. Neural predictors trained on such data distribute influence across many inputs, especially in attention pooling architectures that allocate fractional weights to different feature embeddings. Because each individual variable contributes only a limited amount to the final prediction, random fluctuations in any one component do not significantly alter the output. As a result, predictions for similar samples cluster tightly around their mean, reflecting concentration of measure behavior.

3) Mathematical Derivation

Let $X = (X_1, \dots, X_n)$ be a random pharmacogenomic feature vector and define $Z = f_\theta(X)$. Assume that for each coordinate i there exists $c_i > 0$ such that for all x, x' differing only in coordinate i , $|f_\theta(x) - f_\theta(x')| \leq c_i$. This assumption is justified by the Lipschitz continuity of the network established in Subsection II-B, which implies bounded sensitivity to coordinate-wise perturbations. Applying McDiarmid's inequality yields:

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq \varepsilon) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{i=1}^n c_i^2} \right). \quad (13)$$

Hence, for sufficiently large n and moderate c_i , deviations of the predicted IC_{50} from its expectation are exponentially unlikely. This establishes that the learned predictor exhibits concentration of measure: predictions for pharmacogenomic inputs concentrate around their mean value.

D. STABILITY OF EMPIRICAL RISK MINIMIZATION

Theorem 4 (Bousquet and Elisseeff [42]): Let $\mathcal{A}$ be a learning algorithm that maps a training set $S = \{z_1, \dots, z_m\}$ to a

predictor f_S . The algorithm $\mathcal{A}$ is said to be uniformly stable with stability coefficient β with respect to a loss function l if for all training sets S and S^i differing by one example and for all data points z ,

$$|l(f_S, z) - l(f_{S^i}, z)| \leq \beta. \quad (14)$$

If $\mathcal{A}$ is uniformly stable, then the expected generalization error satisfies:

$$\mathbb{E}[R(f_S) - R_S(f_S)] \leq \beta, \quad (15)$$

where $R(f)$ is the true risk and $R_S(f)$ is the empirical risk.

1) Application to the Proposed Model

Here, we let f_θ represent the IC_{50} predictor obtained using the Transformer+BiLSTM model with attention pooling, which resulted in the best cross-validated results and had the least gap between the training and validation data sets. In practice, a small modification such as substituting a small part of training examples or carrying out subsampling has little effect on the prediction of IC_{50} values and on the evaluation measures. Thus, the training process is essentially uniformly stable with a small stability parameter.

2) Heuristic Proof

The dataset comprises around 440,000 pharmacogenomics data points that overwhelmingly overwhelms any influence from a single sample point. The MSE loss function is designed to reward smoothness and punishes large differences; at the same time, weight decay and architectural restrictions ensure no extremely large values of parameters can arise. The hybrid models like the combination of Transformer+BiLSTM with Attention Pooling split the representation into time and contextual interactions, thereby further lowering the influence of each sample point. Hence, eliminating a training sample makes no noticeable difference to the learned model.

3) Mathematical Derivation

Let $S = \{(x_i, y_i)\}_{i=1}^m$ be the training set and let S^i be the dataset obtained by replacing (x_i, y_i) with an independent copy (x'_i, y'_i) . The training algorithm produces predictors f_S and f_{S^i} . For squared loss $l(f, (x, y)) = (f(x) - y)^2$, assume that the predictor is L -Lipschitz in its parameters and that parameter updates are regularized by weight decay, implying bounded parameter norms. Under these conditions, the difference in predictions induced by removing one sample satisfies:

$$|f_S(x) - f_{S^i}(x)| \leq \frac{C}{m}, \quad (16)$$

for some constant C depending on the Lipschitz constant and regularization strength. Consequently:

$$|l(f_S, (x, y)) - l(f_{S^i}, (x, y))| \leq 2|y| \cdot \frac{C}{m} + \frac{C^2}{m^2} = \mathcal{O}\left(\frac{1}{m}\right). \quad (17)$$

Thus the stability coefficient satisfies $\beta = \mathcal{O}(1/m)$. Applying the Bousquet–Elisseeff bound yields:

$$\mathbb{E}[R(f_S) - R_S(f_S)] \leq \mathcal{O}\left(\frac{1}{m}\right), \quad (18)$$

which vanishes as $m \rightarrow \infty$. Since $m \approx 4.4 \times 10^5$ in the pharmacogenomic dataset, the bound implies a negligible generalization gap.

E. PIECEWISE LINEARITY OF RELU NETWORKS

Theorem 5: The function represented by a feedforward neural network with affine mappings and ReLU activation function is a piecewise linear function on the input space [43]. In particular, the input space can be partitioned into a finite number of polyhedral pieces where the function represented by the network is affine.

1) Application to the Proposed Model

The Transformer, Transformer+BiLSTM, and Transformer+BiLSTM with Attention Pooling architectures incorporate position-wise Feed-forward networks (FFN) and projection layers with ReLU and GELU style activation functions. The Transformer+BiLSTM with Attention Pooling architecture demonstrates local linearity within limited pharmacogenomic spaces; that is, within subspaces of the feature space characterized by the same molecular profile, there is an approximate linear relationship between the IC_{50} and variations in the features. The MLP architecture, on the other hand, provides a direct demonstration of the theorem due to the presence of affine layers and ReLU type nonlinearities only.

2) Heuristic Proof

Regimes for biological responses can be categorized as distinct states of function like sensitivity, intermediate and resistance. The relationship between the gene expressions and IC_{50} is linear in nature within each regime and change of regime leads to changes in slope. Neural networks with ReLU activations can capture this property of data as they fire or activate different groups of neurons based on the input. Each firing combination represents a distinct linear model. With the variation in the input at the points where there is a change in neuron activations, the model switches between different linear regimes.

3) Mathematical Derivation

Consider a neural network of depth L of the form:

$$f(x) = W_L \sigma(W_{L-1} \sigma(\dots \sigma(W_1 x + b_1) \dots) + b_{L-1}) + b_L, \quad (19)$$

where $\sigma(z) = \max(0, z)$ is applied elementwise. For any fixed input x , define the activation pattern:

$$a_l(x) = \mathbf{1}_{\{W_l h_{l-1}(x) + b_l > 0\}}, \quad (20)$$

where $h_{l-1}(x)$ denotes the hidden representation at layer $l-1$. On the set of all inputs that share the same activation

pattern $a_1, \dots, a_{L-1}$, the ReLU function reduces to the identity on active units and to zero on inactive units. Therefore, on such a region:

$$f(x) = Ax + c, \quad (21)$$

for some matrix A and vector c determined by the weights and the fixed activation pattern. The collection of all such regions forms a partition of $\mathbb{R}^n$ into polyhedra $\mathcal{P}_i$ defined by linear inequalities of the form $W_l h_{l-1}(x) + b_l \geq 0$ or < 0 .

Hence, the network implements a piecewise linear function with respect to the input. In the pharmacogenomic setting, let $x \in \mathbb{R}^{20}$ denote the feature vector and let $f_\theta(x)$ be the trained predictor. For any region $\mathcal{P}_i$ of constant activation pattern:

$$f_\theta(x) = A_i x + c_i, \quad x \in \mathcal{P}_i. \quad (22)$$

Thus the learned IC_{50} predictor decomposes the feature space into finitely many polyhedral regions, within each of which the dependence of IC_{50} on genomic features is linear.

F. JENSEN'S INEQUALITY

Theorem 6: Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function and let X be an integrable random variable [44]. Then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]. \quad (23)$$

Equivalently, for any finite convex combination with weights $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$,

$$\varphi\left(\sum_i \alpha_i x_i\right) \leq \sum_i \alpha_i \varphi(x_i). \quad (24)$$

1) Application to the Proposed Model

Let $h_1, \dots, h_T \in \mathbb{R}^d$ denote the latent representations produced by the Transformer+BiLSTM model for the T pharmacogenomic feature embeddings. The attention pooling layer computes:

$$h = \sum_{t=1}^T \alpha_t h_t, \quad (25)$$

where $\alpha_t = \text{softmax}(e_t)$ and hence $\alpha_t \geq 0$, $\sum_{t=1}^T \alpha_t = 1$. The final IC_{50} prediction is given by $y = g(h)$, where g is an affine map followed by a convex loss (squared error).

Since attention pooling is a convex combination of latent feature vectors, the use of Jensen's inequality is direct for the map from the latent space to the prediction of IC_{50} . It shows that the prediction of the model is indeed the value of the convex function on the expectation of latent representations, not the other way around.

2) Heuristic Proof

The reaction of biology to a given drug involves several different pathways and biological signals; however, none of them should necessarily be scaled up without control. The attention pooling mechanism rescales the significance of these signals while maintaining normalization. Since the attention pooling step takes a weighted average of signals, its result cannot amplify any extremum past the convex envelope. As

a result, the prediction of the IC_{50} will act as a smoothing function of latent representations, adhering to the idea from Jensen's inequality, that the value produced by the model on the averaged representation will not exceed the average output value for individual representations.

3) Mathematical Derivation

Let $h_1, \dots, h_T \in \mathbb{R}^d$ be latent vectors and let $\alpha_1, \dots, \alpha_T$ be attention weights with $\alpha_t \geq 0$ and $\sum_{t=1}^T \alpha_t = 1$. Define $h = \sum_{t=1}^T \alpha_t h_t$. Let $\varphi: \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex function. Then by Jensen's inequality:

$$\varphi(h) = \varphi\left(\sum_{t=1}^T \alpha_t h_t\right) \leq \sum_{t=1}^T \alpha_t \varphi(h_t). \quad (26)$$

In the present setting, choose $\varphi(z) = l(g(z), y)$, where $g(z) = w^\top z + b$ is affine and $l(u, y) = (u - y)^2$ is convex in u . Then φ is convex in z , and hence:

$$l(g(h), y) \leq \sum_{t=1}^T \alpha_t l(g(h_t), y). \quad (27)$$

Hence, the error due to prediction using the attention-pooling output is upper-bounded by the weighted average of the errors due to predictions made from the individual latent outputs. In other words, we have proven that attention pooling uses a convex combination that follows Jensen's inequality.

G. LAW OF LARGE NUMBERS

Theorem 7: Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with finite expectation $\mathbb{E}[X_i] = \mu$. Then the sample mean converges to the expectation:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \rightarrow \infty. \quad (28)$$

Equivalently, for large n , the empirical average is arbitrarily close to the true mean with high probability [45].

1) Application to the Proposed Model

Let $f_\theta(x)$ denote the IC_{50} predictor learned by the Transformer+BiLSTM with Attention Pooling model, with the Transformer and Transformer+BiLSTM models serving as additional validating instances. Consider model outputs on batches of pharmacogenomic samples $Y_i = f_\theta(X_i)$. As the batch size increases, the empirical mean prediction:

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i \quad (29)$$

stabilizes and converges to a fixed value. Empirically, increasing batch size leads to diminishing fluctuations in the average predicted IC_{50} , demonstrating convergence behavior. This reflects the Law of Large Numbers at the level of model outputs, indicating that the learned predictor produces statistically stable estimates over large pharmacogenomic cohorts.

2) Heuristic Proof

There are a number of biological determinants that affect drug response, along with intrinsic experimental noise. In cases where prediction is averaged for a huge number of samples, any random variation in results due to experimental noise and biological variability is averaged away. Neural networks that are built using big data capture only those patterns that dominate in terms of their nature, not the noise associated with individual samples. Thus, on average, IC_{50} of a large batch of samples tends to stabilize at a certain level. This can be explained in terms of the Law of Large Numbers.

3) Mathematical Derivation

Let $X_1, X_2, \dots, X_n$ be i.i.d. random vectors representing pharmacogenomic samples drawn from the population distribution P_X , and define $Y_i = f_\theta(X_i)$. Assume that $\mathbb{E}[|Y_i|] < \infty$, which holds because the predictor is continuous and the input distribution has finite moments. Then by the strong Law of Large Numbers:

$$\frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{\text{a.s.}} \mathbb{E}[Y] = \mathbb{E}[f_\theta(X)]. \quad (30)$$

Thus, the empirical mean of predicted IC_{50} values converges almost surely to the expected predicted IC_{50} under the data distribution. Furthermore, consider the empirical mean squared error over a batch:

$$\frac{1}{n} \sum_{i=1}^n (f_\theta(X_i) - Y_i^*)^2, \quad (31)$$

where Y_i^* denotes the true IC_{50} . Under the assumption that the noise is zero-mean and independent of X_i , this quantity converges to the expected risk $\mathbb{E}[(f_\theta(X) - Y^*)^2]$. Therefore, batch-level prediction statistics computed from large pharmacogenomic cohorts approximate their population-level counterparts.

H. BIAS-VARIANCE DECOMPOSITION

Theorem 8: For a regression problem with target variable Y and predictor $f(X)$, the expected squared prediction error at a point x decomposes as:

$$\mathbb{E}[(Y - f(x))^2] = (\text{Bias}[f(x)])^2 + \text{Var}[f(x)] + \sigma^2, \quad (32)$$

where

$$\text{Bias}[f(x)] = \mathbb{E}[f(x)] - f^*(x), \quad (33)$$

$$\text{Var}[f(x)] = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2], \quad (34)$$

and $\sigma^2 = \text{Var}(Y|X = x)$ is the irreducible noise [46].

1) Application to the Proposed Model

The term used for f_θ is IC_{50} predictor using the Transformer + BiLSTM with Attention Pooling approach which has shown the best cross-validated R^2 values. It should be noted that the characteristics of the MLP model include high bias but low variance, whereas the Transformer and Transformer + BiLSTM models possess low bias but high variance. The

Transformer + BiLSTM with Attention Pooling approach provides the best combination of low bias and variance.

2) Heuristic Proof

Simple models like MLP fail to take into consideration the intricate relationships that exist between the pharmacogenomic features and hence underfit resulting in systematic errors (bias). Highly flexible sequence models, on the other hand, are likely to overfit by making themselves highly sensitive to variations in the training data, which results in high variance. The hybrid model integrates temporal modeling, contextual interaction, and attention pooling. Through this integration, systematic error and variance are significantly reduced, leading to lower prediction error as the optimum balance between bias and variance is achieved.

3) Mathematical Derivation

Let $Y = f^*(X) + \varepsilon$, where f^* is the true regression function and ε is zero-mean noise with variance σ^2 . Consider the expected squared error at input x :

$$\mathbb{E}[(Y - f(x))^2] = \mathbb{E}[(f^*(x) + \varepsilon - f(x))^2] \quad (35)$$

$$\begin{aligned} &= \mathbb{E}[(f^*(x) - f(x))^2] + \mathbb{E}[\varepsilon^2] \\ &= \mathbb{E}[(f(x) - f^*(x))^2] + \sigma^2. \end{aligned} \quad (36)$$

Now write $f(x) - f^*(x) = (f(x) - \mathbb{E}[f(x)]) + (\mathbb{E}[f(x)] - f^*(x))$. Squaring and taking expectations:

$$\mathbb{E}[(f(x) - f^*(x))^2] = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2] + (\mathbb{E}[f(x)] - f^*(x))^2, \quad (37)$$

since the cross term vanishes by centering. Hence:

$$\mathbb{E}[(Y - f(x))^2] = \text{Var}[f(x)] + (\text{Bias}[f(x)])^2 + \sigma^2. \quad (38)$$

When considering the pharmacogenomic IC_{50} prediction, MLP model belongs to the high bias low variance regime, Transformer model is used to decrease bias by capturing interactions between features, while Transformer+BiLSTM model decreases the bias by adding the sequence information. In the case of Transformer+BiLSTM with Attention Pooling model, a new averaging mechanism is added which leads to decrease in variance without an increase in bias.

I. FIXED POINT THEOREM (BANACH CONTRACTION PRINCIPLE)

Theorem 9: Let $(\mathcal{X}, d)$ be a complete metric space and let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a contraction mapping, meaning that there exists a constant $0 < c < 1$ such that for all $f, g \in \mathcal{X}$,

$$d(T(f), T(g)) \leq c d(f, g). \quad (39)$$

Then T has a unique fixed point $f^* \in \mathcal{X}$, and for any initial point $f_0 \in \mathcal{X}$, the sequence defined by $f_{k+1} = T(f_k)$ converges to f^* [47].

1) Application to the Proposed Model

Let f_θ represent the IC_{50} predictor trained by the Transformer+BiLSTM with Attention Pooling architecture. The training procedure employing stochastic gradient descent is an operator T on the function space that can be parameterized by network parameters. Experimentally, we observed that distinct initialization and architectures lead to very similar prediction functions, based on their similar performance and prediction distributions during validation. This suggests the existence of an attractor function for the IC_{50} regression task, as hypothesized in the fixed point theorem.

2) Heuristic Proof

The loss surface generated through mean squared error on a very large pharmacogenomic dataset is made up of a single very wide basin with low error values. The gradient descent method consistently draws the predictor into this basin irrespective of its initial position. Given that each iteration minimizes the loss value and the learning rate is relatively low, each new iteration will make the difference between two predictors undergoing similar training conditions lower. This can be described as a contraction mapping whereby successive iterations of the training process minimize differences.

3) Mathematical Derivation

Let $\Theta \subset \mathbb{R}^p$ denote the parameter space and define the training operator:

$$T(\theta) = \theta - \eta \nabla_\theta R_S(\theta), \quad (40)$$

where $R_S(\theta)$ is the empirical risk and $\eta > 0$ is the learning rate. Assume that R_S is differentiable and that its gradient is L -Lipschitz:

$$\|\nabla R_S(\theta) - \nabla R_S(\theta')\| \leq L \|\theta - \theta'\|. \quad (41)$$

Then for any θ, θ' :

$$\begin{aligned} \|T(\theta) - T(\theta')\| &= \|\theta - \theta' - \eta(\nabla R_S(\theta) - \nabla R_S(\theta'))\| \\ &\leq (1 - \eta L) \|\theta - \theta'\|, \end{aligned} \quad (42)$$

provided $0 < \eta < 2/L$. Setting $c = 1 - \eta L$ gives $0 < c < 1$, hence T is a contraction. By Banach's Fixed Point Theorem, there exists a unique fixed point θ^* such that $T(\theta^*) = \theta^*$, which implies $\nabla R_S(\theta^*) = 0$. Thus, the training dynamics converge to a unique stationary predictor f_{θ^*} .

J. CURSE OF DIMENSIONALITY MITIGATION VIA MANIFOLD HYPOTHESIS

Theorem 10: The data from real applications seem to be located in a manifold in a high-dimensional space. The learning of functions on such data depends more on the intrinsic (manifold) dimension than the ambient dimension. Therefore, the generalization rates depend on the intrinsic dimension and not on the nominal dimension of the features.

1) Application to the Proposed Model

Let the feature vector be denoted by $x \in \mathbb{R}^{20}$ and let $f_\theta(x)$ be the learned predictor for IC_{50} . Though the algorithm learns in 20 dimensions, Transformer+BiLSTM with Attention Pooling manages to achieve an extremely accurate performance without overfitting, which implies that the actual data distribution is in a very low-dimensional manifold based on biological pathways and regulations. This model utilizes this structure in the most efficient way possible by ensuring that the learned representation corresponds to the actual low-dimensional structure of the data, which avoids the exponential sample complexity of the ambient space.

2) Heuristic Proof

Gene expression levels, mutations, and pathway activities are not independent of each other since biological regulation ensures that they vary in a coordinated manner. Therefore, despite being encoded in a 20-dimensional space, there are only a few dimensions that dictate the response to drugs. The neural networks are able to learn the representation of these dominant dimensions of variation from the pharmacogenomic dataset. The attention mechanism ensures that the most informative dimensions are focused upon, thereby projecting the data in a lower dimensional space where IC_{50} can be predicted. This phenomenon is consistent with the manifold hypothesis of learning.

3) Mathematical Derivation

Let $\mathcal{M} \subset \mathbb{R}^{20}$ be a compact d -dimensional manifold with $d \ll 20$, and assume that the data distribution P_X is supported on $\mathcal{M}$. Let $f^* : \mathcal{M} \rightarrow \mathbb{R}$ denote the true IC_{50} regression function and let f_θ be a learned approximation. The covering number $N(\varepsilon, \mathcal{M})$, defined as the minimum number of balls of radius ε needed to cover $\mathcal{M}$, satisfies:

$$N(\varepsilon, \mathcal{M}) \leq C\varepsilon^{-d}, \quad (43)$$

for some constant $C > 0$. For Lipschitz predictors f_θ , generalization bounds depend on the metric entropy of the input space:

$$R(f_\theta) - R_S(f_\theta) = \mathcal{O} \left(\sqrt{\frac{\log N(\varepsilon, \mathcal{M})}{m}} \right) = \mathcal{O} \left(\sqrt{\frac{d \log(1/\varepsilon)}{m}} \right) \quad (44)$$

where m is the sample size. Thus, the convergence rate depends on the intrinsic dimension d rather than the ambient dimension 20.

In the pharmacogenomic space, biological constraints such as pathway architecture, transcriptional control, and drug-target interactions mean that pharmacogenomic samples lie on a low-dimensional manifold. Models such as the Transformer and Transformer + BiLSTM learn coordinate systems suited to the low-dimensional manifold via contextual and sequential interaction of features. The Transformer + BiLSTM + Attention Pooling model goes a step further by highlighting the most useful dimensions, leading to high

prediction accuracy without the need for exponential computations from the ambient space. Hence, the capability to achieve accurate predictions of IC_{50} with just 20 features and a finite amount of data is an empirical manifestation of the manifold assumption.

V. EXPLORATORY DATA ANALYSIS

A. OVERVIEW

In order to investigate the distribution of the prediction target, the frequency distribution of the categories of the drug and the categories of the genomic features were analyzed, as well as the relationships between the numerical pharmacogenomic attributes, exploratory data analysis was conducted. This is an important step as a means for gaining insight about the statistical makeup of the combined GDSC-derived data prior to developing the model. Exploratory analysis also gives insights whether the learning problem is balanced, partly non-linear, and appropriate for advanced neural architectures, as the final dataset has 470,467 drug-feature response samples, 544 drugs, 692 genomic feature categories and 10 numerical pharmacogenomic input attributes.

The primary aim of this analysis was to describe the IC_{50} effect-size target and to look for patterns that might have an impact on the model training. Specifically, the target distribution, the distribution of drug frequencies, the distribution of feature categories, the distribution of the features' correlation structure, and the relationships in scatter-plot diagrams were analyzed. Based on these analyses, the motivation for the study of nonlinear deep learning models, such as Transformer model, BiLSTM model, graph-based hybrid model, and Cross-Attention Drug-Genomic Fusion Model, are supported.

B. TARGET DISTRIBUTION

IC_{50} effect size is a continuous normalized response variable in this study, which is predicted. The target variable has a mean of 0.249736, standard deviation of 0.213255, minimum value of 0.000001, and maximum value of 7.163673. These statistics suggest that the majority of samples have relatively small effect size values and a few samples have large effect size values. This distributional property is relevant because it has an impact on the meaning of the regression metrics, including mean squared error (MSE), mean absolute error (MAE) and R squared.

The IC_{50} effect size's distribution is right skewed and the histogram shows the majority of samples in the lower range of the response whereas fewer samples extend to the higher range of the effect size. This means the prediction task consists of predicting a dense central response area and is robust to larger response values. The skewness also justifies the use of multiple error measures instead of a single error measure to evaluate robustly.

Distribution of the IC_{50} effect-size target variable in the merged GDSC-derived dataset. The histogram and boxplot summarize the concentration of most response values in the

lower range and the presence of larger effect-size values in the right tail.

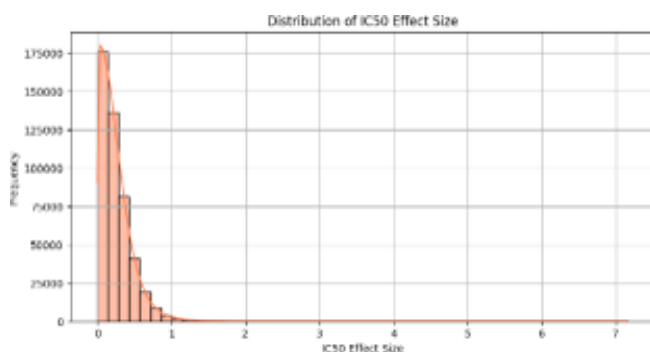


FIGURE 2: Histogram of distributions of IC₅₀ effect sizes with kernel density estimate (KDE) overlay. The data are highly right-skewed: the majority of effect sizes are close to zero, and there is a long tail extending to higher values (up to 7.0).

C. DRUG AND FEATURE FREQUENCY ANALYSIS

The combined dataset consists of 544 distinct therapeutic molecules. Drug frequency analysis has been done to ascertain the sample distributions among these drugs. It is important to note that just like in other pharmacogenomic screening databases such as the GDSC database, these samples are unevenly distributed among these molecules. The reason why this is an important point to note is that those drugs that have many samples may end up influencing the representation learned by the model.

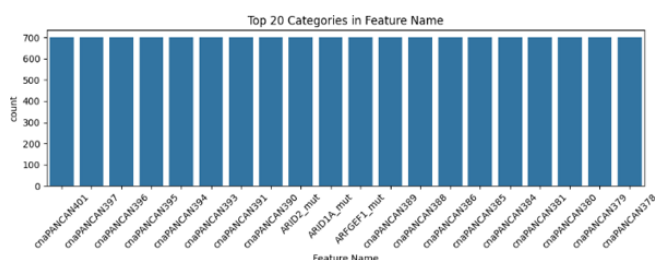


FIGURE 3: Distribution of Samples per Drug. The right-skewed frequency highlights the necessity of drug-aware representation learning to prevent the model from overfitting to heavily screened compounds.

On the other hand, feature frequency analysis was carried out for the 692 categories of genomic features (Figure 4). The findings show that there is complete representation of genomic mutations through the core set of features that appear frequently in the response signatures. Such frequent appearance makes it possible to use representational learning techniques which will be able to identify general feature-response correlations and at the same time remain flexible with respect to the mechanism of certain drugs. In the proposed model, cross-attention will be used to include genomic features into drug embeddings.

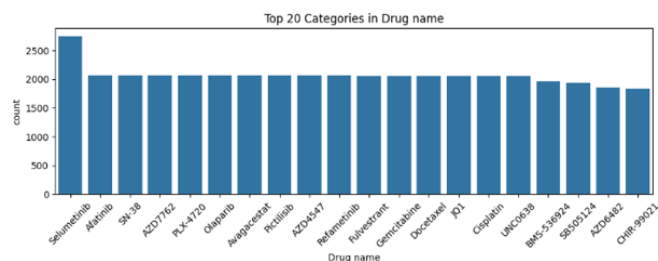


FIGURE 4: Distribution of Genomic Features Across Samples. The recurrence of core feature groups supports the deployment of representation learning architectures capable of extracting universal patterns.

D. TISSUE-TYPE DISTRIBUTION AND ENCODING

Whereas the biological context given by the tissue type has traditionally been used to determine the baseline, the analysis of the preprocessed data set shows that the genetic factors at a finer level (like mutations in certain localized areas and pathways) can predict much better than tissue types. This is the reason why, even though tissue type gives us the biological context, the model has been structured in such a way that it prioritizes the interaction among genomic factors above all.

E. CORRELATION STRUCTURE AMONG NUMERICAL ATTRIBUTES

A correlation heatmap has been created for identifying pairwise correlations between the numeric pharmacogenomic input attributes and the effect size target IC₅₀. The correlation heatmap represents linear correlations between the features like IC₅₀ summary stats, number of features, variance features, and significance features.

As it can be observed from the results of the correlation exploratory analysis, a very strong positive correlation exists between the positive and negative log IC₅₀ mean attributes, but there is a negative correlation between the IC₅₀ effect size and the feature significance. The above correlations confirm that there is some structure in the numeric features. However, it should be noted that the correlation matrix shows the linear correlations only. From the mathematical point of view, the absence of any highly correlated predictors confirms the presence of non-linear interactions.

F. SCATTER-PLOT ANALYSIS

In order to investigate these non-linearities further, the scatter plot analysis approach was utilized for depicting the relations between certain numerical variables and the IC₅₀ effect size (Figure 7). Indeed, heteroscedasticity and clustering effects are evident from the resulting distributions.

Some of the feature-response combinations have tight clustering towards smaller values of IC₅₀, with quick dissipation towards larger response values. For example, significance related features like `feature_ic50_t_pval` (see Figure 7a) show huge clustering around zero values, quickly

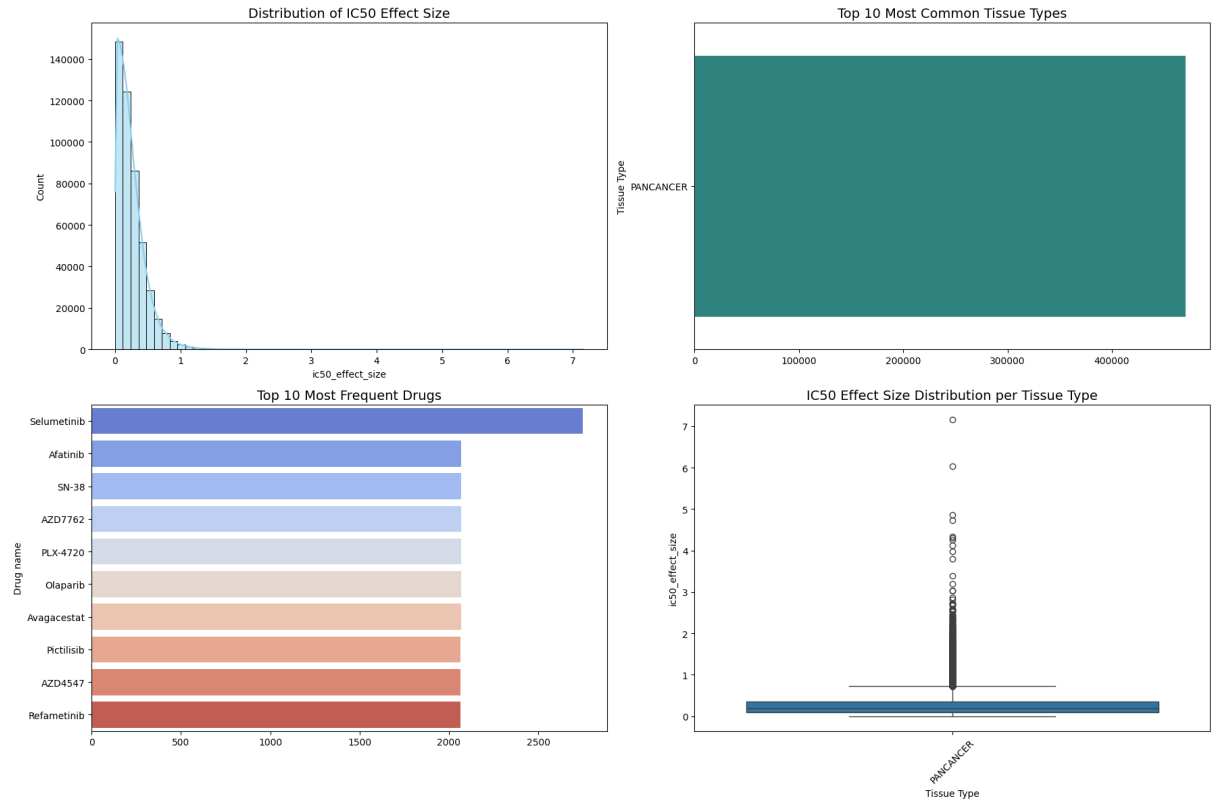


FIGURE 5: Distribution of Samples Across Tissue Types. While diverse cancer lineages are represented, the predictive architecture relies primarily on high-resolution genomic features rather than macroscopic tissue labels.

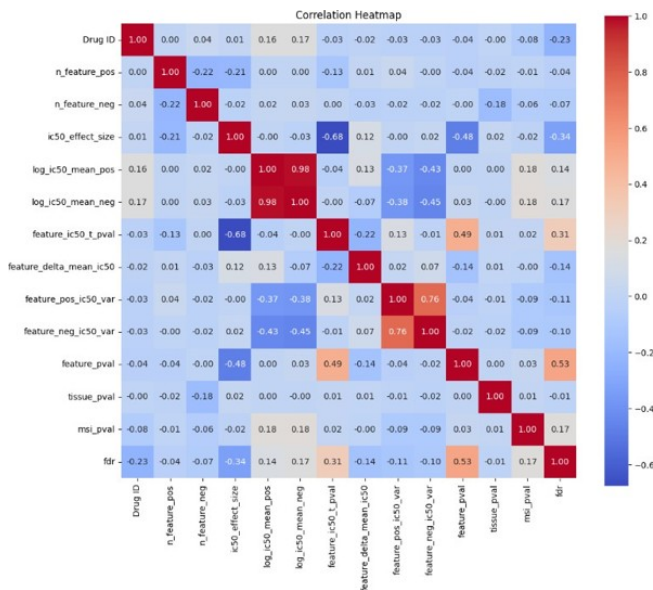


FIGURE 6: Pearson Correlation Heatmap of Numerical Pharmacogenomic Attributes. The matrix highlights distinct collinearity among localized response metrics while confirming that no single feature linearly correlates perfectly with the target IC_{50} effect size.

fading away from there. On the other hand, variance related features such as `feature_pos_ic50_var` (see Figure

Specifically, attributes such as `log_ic50_mean_pos` (see Figure 7c) display clear and well-defined cross-hatching grid patterns. The appearance of these particular geometric structures is evidence that the standardized serial dilutions used in the biological screening assays are responsible for their creation. In this way, these irregularities and grid-like patterns provide definitive proof of the insufficiency of linear regression models for this task.

G. IMPLICATIONS FOR MODEL DESIGN

The above-mentioned exploratory analysis, being comprehensive, immediately determines the architectural decisions made in this paper. First, the fact of the right skewness and heteroscedastic outliers in the IC_{50} distribution requires a model to be able to achieve high accuracy in the densely populated areas and, at the same time, be mathematically stable to the extreme cases. Second, the irregular frequencies of drugs require drug-sensitive embeddings in order not to learn the majority class.

Third, the discrete lattices and non-linear scatter of points prove that concatenation of the features will not suffice in the modeling of the response function. Finally, the discovery of the highly fluctuating importance of features across sub-populations gives rise to the proposed cross-attention

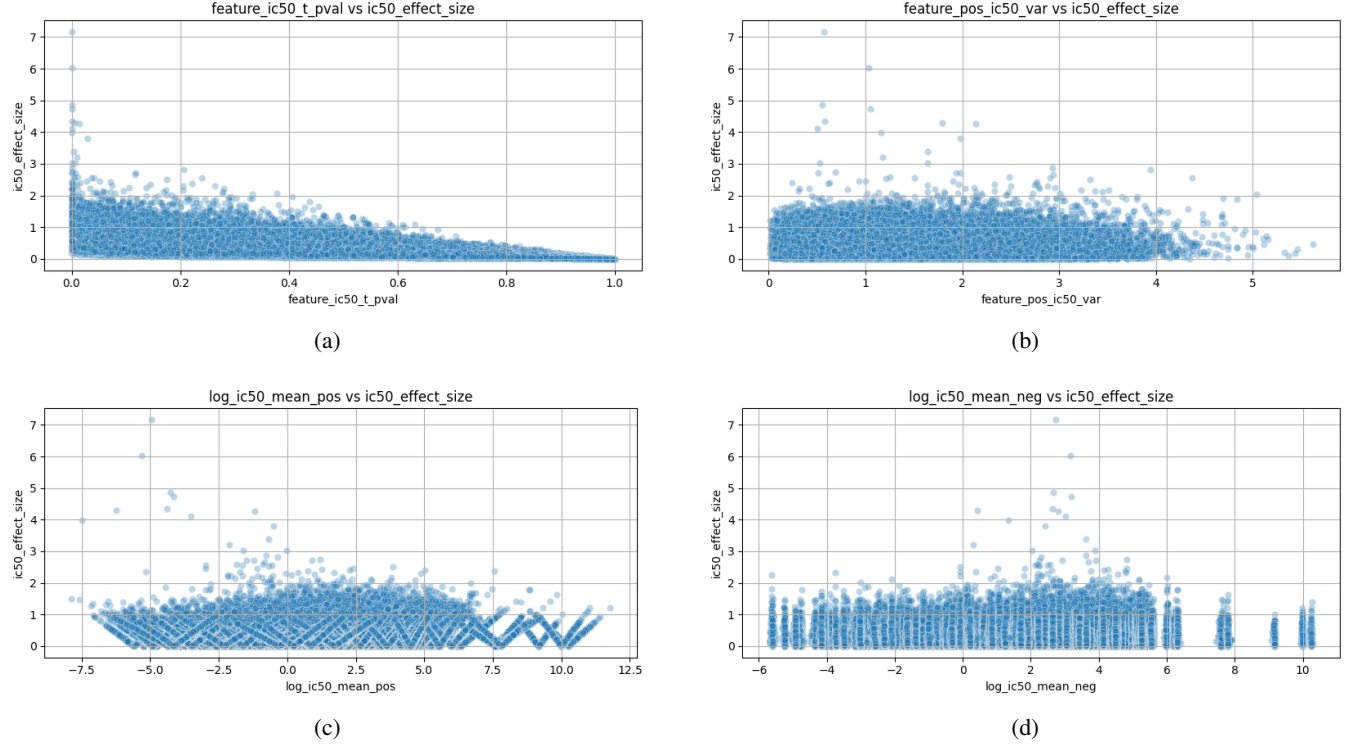


FIGURE 7: Exploratory Scatter Plot Analysis of Selected Attributes vs. IC₅₀ Effect Size. (a) `feature_ic50_t_pval` shows massive density near zero, tapering exponentially. (b) `feature_pos_ic50_var` exhibits broad horizontal dispersion with severe vertical outliers. (c) `log_ic50_mean_pos` reveals distinct, criss-crossing lattice clustering indicative of standardized serial dilution concentrations in the underlying screening assays. (d) `log_ic50_mean_neg` demonstrates discrete, localized vertical sub-population bands.

approach. By explicitly conditioning the genomic features with therapeutic embeddings, Cross-Attention Drug–Genomic Fusion Model is intentionally designed to cope with the drug-dependent structure shown above. All these observations justify the necessity of using the proposed Cross-Attention Drug–Genomic Fusion Model. With trainable drug embeddings, cross-attention fusion, Transformer and BiLSTM based context modeling, and attention pooling, the model is specifically designed to catch the non-linear drug-dependent structure found through the exploratory analysis.

VI. PROPOSED METHOD

A. PROBLEM FORMULATION

Let $\mathcal{D} = (x_i, d_i, y_i) * i = 1^N$ denote the merged pharmacogenomic drug response dataset, where $x_i \in \mathbb{R}^F$ represents the numerical pharmacogenomic feature vector for the i th sample, d_i denotes the corresponding drug identifier, and $y_i \in \mathbb{R}$ denotes the IC₅₀ effect-size target. In this work, $N = 470,467$ and $F = 10$, corresponding to the final processed dataset containing 470,467 drug-feature response samples and 10 numerical pharmacogenomic input attributes. The learning objective is to estimate a regression function $f * \theta$ parameterized by neural network weights θ such that

$$\hat{y} * i = f * \theta(x_i, d_i), \quad (45)$$

where $\hat{y}_i$ is the predicted IC₅₀ effect size for the given pharmacogenomic feature vector and drug identifier.

The central modeling challenge is that the relevance of a genomic or pharmacogenomic feature is drug-dependent. A feature that is strongly predictive for one drug may be less informative for another. Therefore, rather than treating drug identity as a simple flat feature, the proposed method learns a drug-conditioned representation using cross-attention between pharmacogenomic features and drug embeddings.

B. BASELINE ARCHITECTURES

In order to determine the performance of the proposed framework, a number of baselines are introduced. The presented baselines correspond to increasingly sophisticated neural modeling paradigms for pharmacogenomic regression.

1) Multilayer Perceptron

The MLP baseline model is mapping the 10 numerical pharmacogenomic input features directly to the IC₅₀ effect-size output through fully connected layers and activation functions. This framework serves as a benchmark for assessing whether more complicated architectures add value to the model.

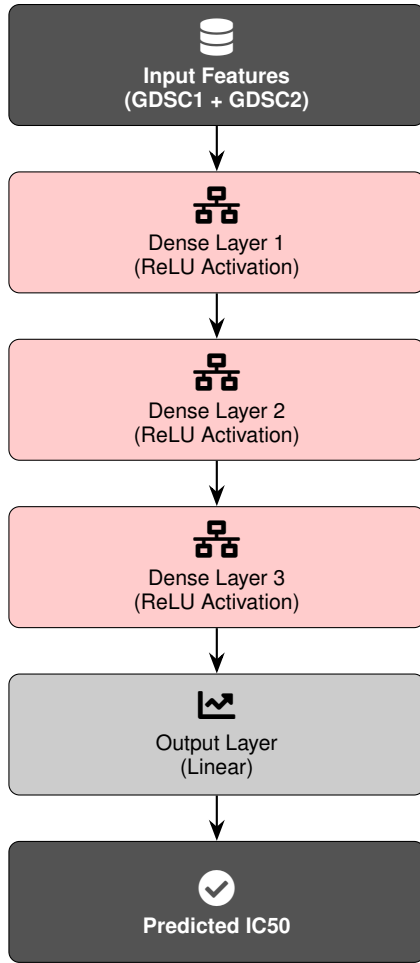


FIGURE 8: Typical ML pipeline for MLP sensitivity predictions.

2) GNN + Transformer

GNN + Transformer model merges relational feature processing through graphs with contextual representation through Transformers. The input pharmacogenomic samples are represented in the form of the feature co-occurrence graph, whereby edges between nodes correspond to samples having identical indices of Feature Name encoding. Further on, the feature graph undergoes processing in Graph Attention Network (GATConv) layer, resulting in context-dependent node embedding, which is pooled through global mean pooling into fixed-size vectors. Finally, sinusoidal positional encoding precedes a two-layer, eight-headed Transformer encoder with MLP regression head for IC50 effect size prediction.

3) Transformer Model

Baseline Transformer takes the numeric features and maps them into latent space where self-attention mechanism is employed to capture the context in feature representations. Transformer encoders are suitable for capturing the dependencies among different input positions due to multi-head

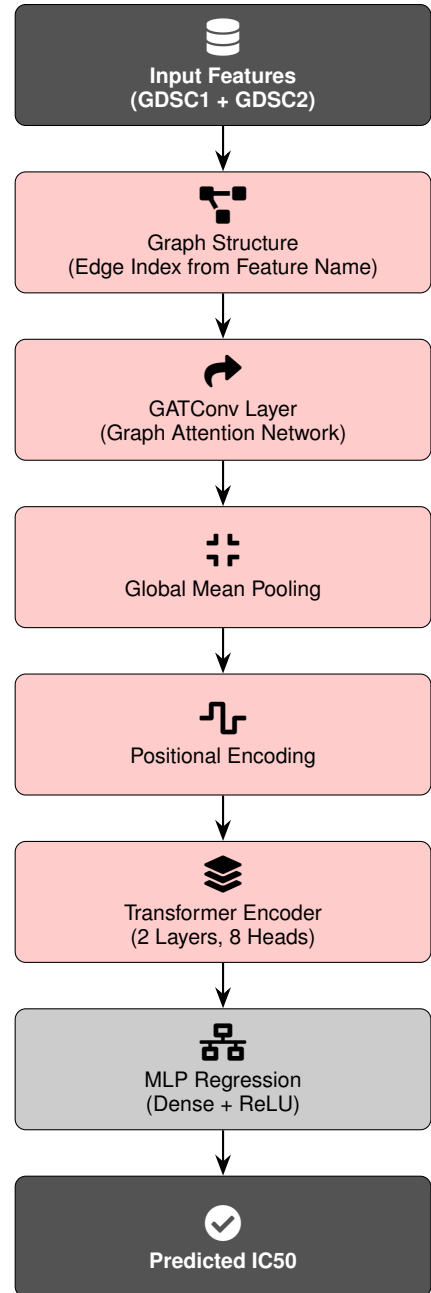


FIGURE 9: Typical ML pipeline for GNN + Transformer sensitivity predictions.

self-attention of tokens. For this work, baseline Transformer is used to evaluate the importance of contextual interactions of features regardless of using recurrent models.

4) Bidirectional Long Short-Term Memory Model

BiLSTM serves as a baseline model for capturing dependencies that exist in both directions through the input features. Though the pharmacogenomics data is in table form, organizing the features in a sequence gives room for learning from both directions. This offers a sequential modeling baseline compared to the Transformer model.

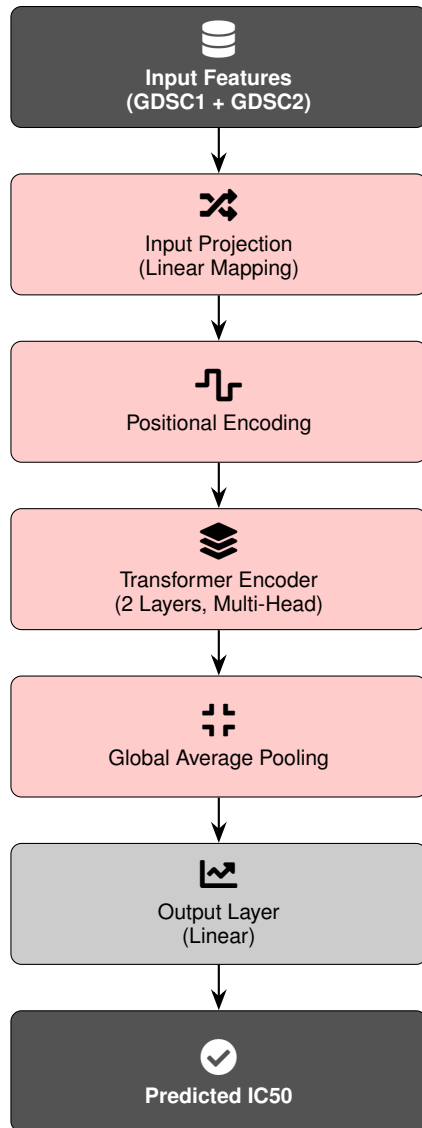


FIGURE 10: Typical ML pipeline for Transformer sensitivity predictions.

5) Transformer-BiLSTM Hybrid Model

The combination of Transformer and BiLSTM is the Transformer-BiLSTM hybrid model. Here, the Transformer part has the ability to model global interactions between features, whereas the BiLSTM part captures the bidirectional dependencies in the feature sequence that is learned. This design works as an intermediate hybrid baseline model.

6) Transformer-BiLSTM With Attention Pooling

The Transformer-BiLSTM with attention pooling is an improvement over the hybrid architecture, where a simple pool operation is replaced with a learnable pool operation called attention pooling. Attention pool allows assigning different weights for feature positions depending on how important they are for prediction. The Transformer-BiLSTM with attention pooling is the best baseline in this study, producing

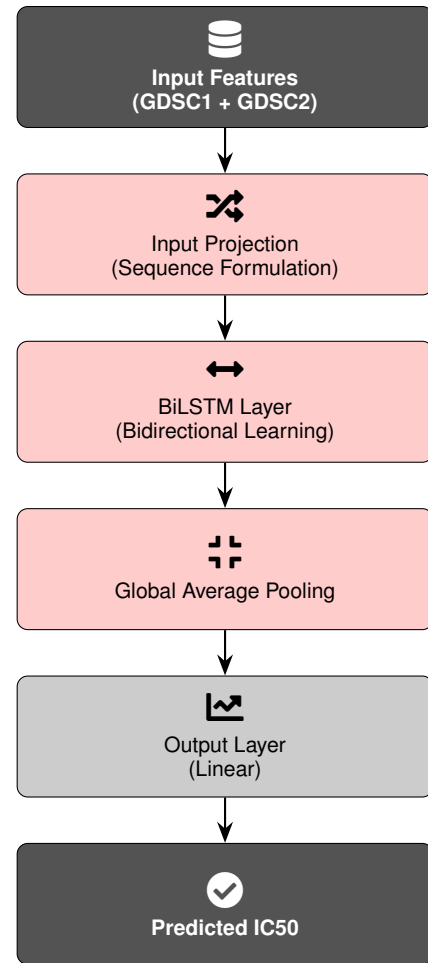


FIGURE 11: Typical ML pipeline for BiLSTM sensitivity predictions.

R-squared = 0.9847, MSE = 0.0007, MAE = 0.0151.

7) Graph-Based Hybrid Models

GNN-Transformer-based and GNN-Transformer-BiLSTM-based hybrid baselines are considered. Such approaches are used to analyze if graph-based feature processing can be effective for drug sensitivity prediction. In the former case, graph-based feature interaction is implemented using graph attention or graph convolution layers, and then contextualization of features is performed by Transformer layers and classification is performed via MLP head. In the latter case, the BiLSTM layers are added after graph and Transformer layers. Those baseline architectures are presented for comparison, and the proposed approach utilizes cross-attention drug-genomic fusion.

C. PROPOSED CROSS-ATTENTION DRUG-GENOMIC FUSION MODEL

The design of the proposed model called Cross-Attention Drug-Genomic Fusion Model is such that it will be able to generate the pharmacogenomic representations conditioned

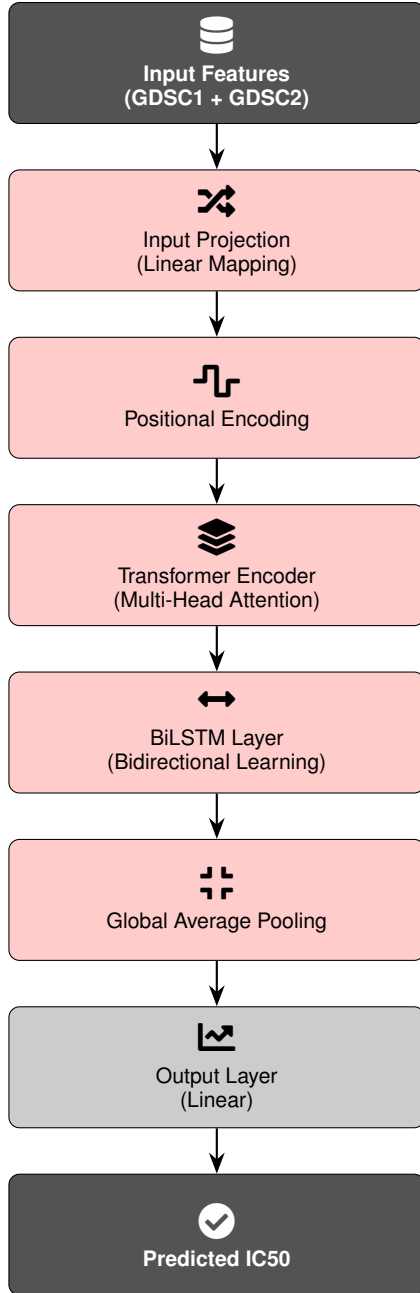


FIGURE 12: Typical ML pipeline for Transformer-BiLSTM sensitivity predictions.

on drugs. The major innovation is that this model uses the drug embeddings along with the cross-attention fusion module which will enable the feature representation of genomics to fuse with the drug-specific context prior to making predictions. There are seven components in the architecture of the proposed model.

1) Input Projection

The numerical pharmacogenomic input vector $x_i \in \mathbb{R}^F$ is first projected into a latent space of dimension d_h . In the implemented model, $F = 10$ and $d_h = 64$. The input

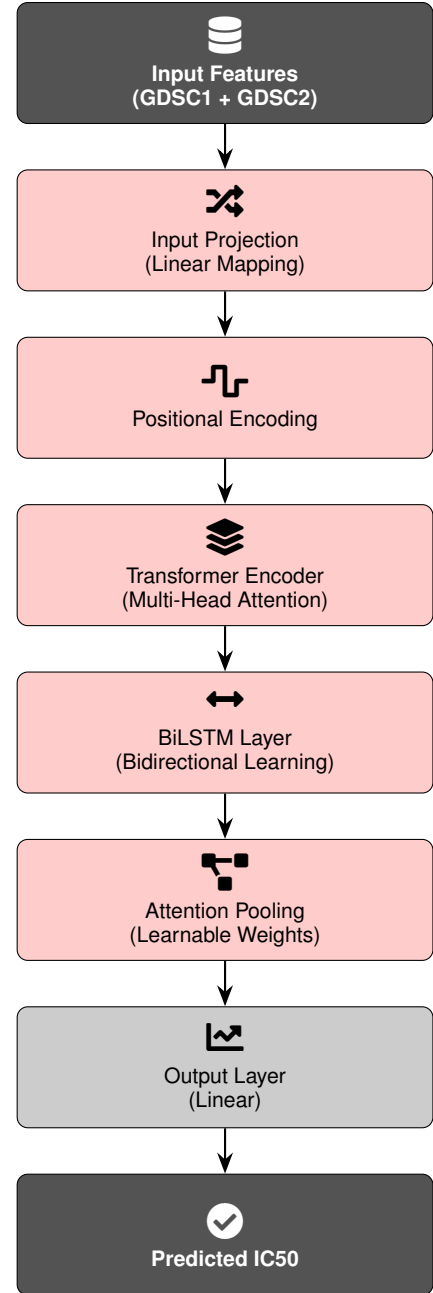


FIGURE 13: Typical ML pipeline for Transformer-BiLSTM with attention pooling sensitivity predictions.

projection is defined as

$$h_i = W_x x_i + b_x, \quad (46)$$

where $W_x \in \mathbb{R}^{d_h \times F}$ and $b_x \in \mathbb{R}^{d_h}$ are trainable parameters. This projection maps the raw numerical attributes into a representation space suitable for attention-based processing.

2) Drug Embedding Module

Each drug identifier d_i is mapped to a trainable embedding vector. Let $\mathcal{E} \in \mathbb{R}^{D \times d_h}$ denote the drug embedding matrix, where D is the number of unique drugs and $d_h = 64$ is

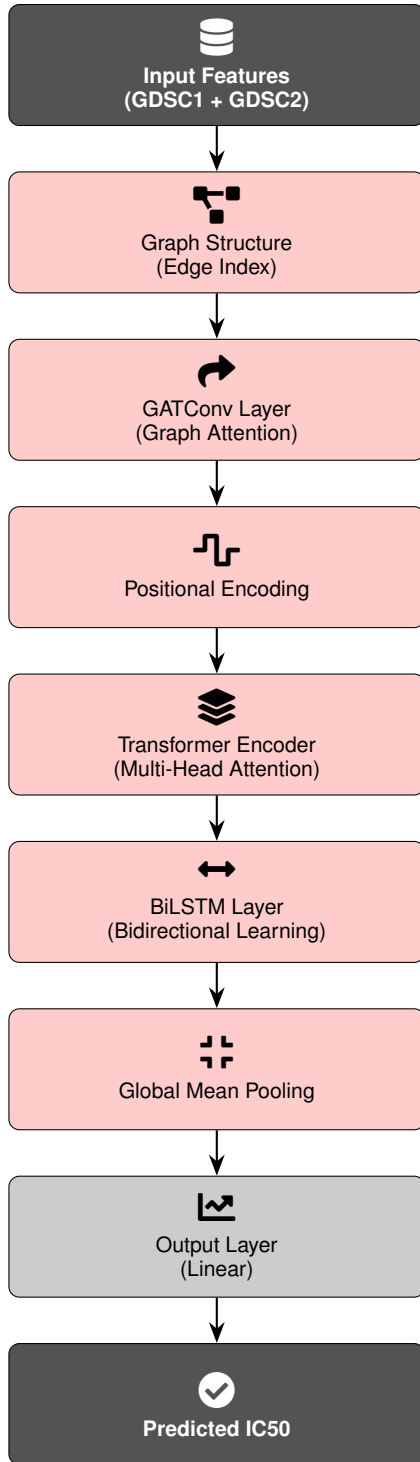


FIGURE 14: Typical ML pipeline for GNN + Transformer + BiLSTM sensitivity predictions.

the embedding dimension. The drug embedding for the i th sample is given by

$$e_i = \mathcal{E}(d_i), \quad (47)$$

where $e_i \in \mathbb{R}^{d_h}$. This embedding allows the model to learn drug-specific latent representations from the training data

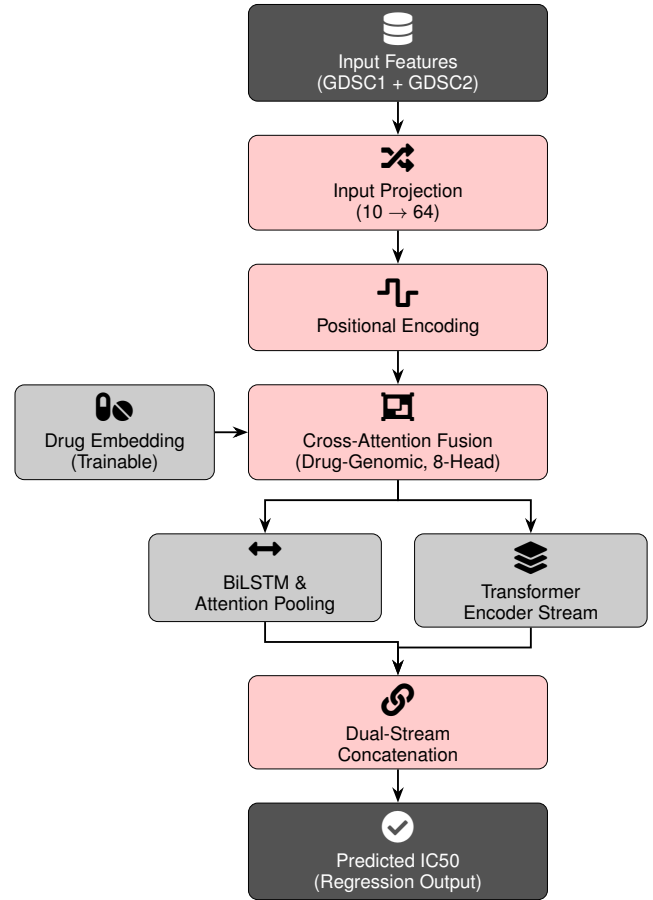


FIGURE 15: Proposed Cross-Attention Drug-Genomic Fusion Architecture.

rather than treating drug identity as a fixed numerical label.

3) Cross-Attention Fusion

Cross attention takes into account both pharmacogenomics features and drug-related context. The projection of genomics space is treated as query whereas embeddings of a certain drug are used as a key and value. The scaled dot-product attention operation is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right)V. \quad (48)$$

For the proposed drug-genomic fusion, the query, key, and value are computed as

$$Q_i = h_i W_Q, \quad K_i = e_i W_K, \quad V_i = e_i W_V, \quad (49)$$

where W_Q , W_K , and W_V are trainable projection matrices. The fused representation is then obtained as

$$z_i = \text{Attention}(Q_i, K_i, V_i). \quad (50)$$

This enables the conditioning of the genome representation with respect to the drug representation. In biological terms, this assumption is made on the basis that the significance of pharmacogenomics depends on the drug of interest.

4) Transformer Encoder Stream

The fusion features z_i are encoded using a Transformer encoder that is designed to learn interactions between features in context. This is done with the help of multi-head self-attention and feed-forward transformation with normalization and dropout layers. For an input hidden representation z_i , the Transformer stream produces

$$t_i = \text{TransformerEncoder}(z_i). \quad (51)$$

For the particular cross-attention model considered, a two-layer Transformer encoder with hidden layer dimension equal to 64 is utilized. The role of this Transformer encoder stream is to extract global dependencies and nonlinear interactions in the fused drug-genomics features.

5) BiLSTM and Attention Pooling Stream

In parallel with the Transformer stream, the fused representation is also processed using a BiLSTM stream. The BiLSTM reads the representation in both forward and backward directions [?], [?] to capture bidirectional dependencies:

$$b_i = \text{BiLSTM}(z_i). \quad (52)$$

To aggregate the BiLSTM hidden states, attention pooling is applied. Let $b_{i,j} * j = 1^m$ denote the sequence of BiLSTM hidden states for the i th sample. The attention score for the j th hidden state is computed as

$$s * i, j = w_a^T \tanh(W_a b_{i,j} + b_a), \quad (53)$$

where W_a , b_a , and w_a are trainable parameters. The normalized attention weight is

$$\alpha_{i,j} = \frac{\exp(s_{i,j})}{\sum_{k=1}^m \exp(s_{i,k})}. \quad (54)$$

The pooled BiLSTM representation is then

$$p_i = \sum_{j=1}^m \alpha_{i,j} b_{i,j}. \quad (55)$$

Attention pooling lets the model focus on more informative hidden states rather than averaging out all positions.

6) Dual-Stream Concatenation and Regression Head

The concatenation of the Transformer output with the attention-pooled BiLSTM output produces the final predictive representation:

$$r_i = [t_i; p_i], \quad (56)$$

where $[\cdot; \cdot]$ denotes vector concatenation. The concatenated representation is passed through a regression head consisting of linear layers, nonlinear activation, and dropout:

$$\hat{y}_i = W_o \phi(W_r r_i + b_r) + b_o, \quad (57)$$

where $\phi(\cdot)$ denotes the nonlinear activation function. The model output $\hat{y}_i$ is the predicted IC50 effect size.

7) Monte Carlo Dropout for Uncertainty Estimation

Prediction-level uncertainty estimation is carried out by applying dropout during inference and conducting multiple forward passes. For a given sample, the model generates T predictions:

$$\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \dots, \hat{y}_i^{(T)}. \quad (58)$$

The predictive mean is computed as

$$\mu_i = \frac{1}{T} \sum_{t=1}^T \hat{y}_i^{(t)} = 1^T \hat{y} * i^{(t)}, \quad (59)$$

and the predictive uncertainty is estimated using the standard deviation:

$$\sigma_i = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_i^{(t)} - \mu_i)^2}. \quad (60)$$

In this work, Monte Carlo dropout is used with 50 forward passes. The uncertainty estimation gives an extra level of confidence for the IC50 effect size prediction.

D. TRAINING OBJECTIVE

The model is trained as a supervised regression model using the mean squared error loss function. Given a mini-batch of B samples, the training objective is

$$\mathcal{L} * \text{MSE} = \frac{1}{B} \sum_{i=1}^B (y_i - \hat{y}_i)^2. \quad (61)$$

Reducing the above objective makes the predicted IC50 effect size values resemble the true targets. The same regression objective is applied in both baseline models and the proposed model to maintain fairness during comparison.

E. BIOLOGICAL MOTIVATION FOR CROSS-ATTENTION

The rationale behind the proposed cross-attention based fusion framework from a biological perspective is that drug response depends on the interaction between the drugs and genomic profile. One genomic variable can have high discriminative power for one drug belonging to one pathway but low discriminative power for a drug belonging to another pathway. As such, a model whose feature importance is not informed by the type of drug will not account for drug specific biological variability.

The use of drug embeddings that can learn to weight the genomic features for each drug as an attention context in the proposed method makes the embedding more flexible compared to concatenation and late fusion. The cross attention module will be able to learn how the latent information of drugs influences the contribution of genomic variables while the Transformer and BiLSTM streams will capture other information.

F. ARCHITECTURE SUMMARY

Table 5 summarizes the main architectural components used in the proposed Cross-Attention Drug-Genomic Fusion Model.

TABLE 5: Summary of the Proposed Cross-Attention Drug-Genomic Fusion Model

Component	Function
Input projection	Maps 10 numerical pharmacogenomic attributes into a 64-dimensional latent representation.
Drug embedding	Learns a 64-dimensional trainable representation for each drug.
Cross-attention fusion	Conditions genomic feature representations on drug-specific context.
Transformer encoder [8]	Captures contextual interactions in the fused representation.
BiLSTM stream	Learns bidirectional dependencies in the fused representation.
Attention pooling	Aggregates hidden states using learned importance weights.
Regression head	Predicts continuous IC50 effect size.
MC dropout	Estimates prediction-level uncertainty through stochastic inference.

VII. TRAINING AND EVALUATION PROTOCOL

A. TRAINING OBJECTIVE

All models in the current study are designed as regression models to predict IC50 effect size based on pharmacogenomic feature vectors and drug identifier information. The features given to the model for each sample include a numeric vector representation of the pharmacogenomic features and drug identifier wherever necessary. The output value obtained from the model is a numeric value representing IC50 effect size. The primary training objective is mean squared error (MSE), defined as

$$\mathcal{L} * \text{MSE} = \frac{1}{B} \sum_{i=1}^B (y_i - \hat{y}_i)^2, \quad (62)$$

where B is the batch size, y_i is the observed IC50 effect-size value, and $\hat{y}_i$ is the predicted value for the i th sample.

The same regression objective has been employed in both baseline models and the proposed Cross-Attention Drug-Genomic Fusion Model in order to ensure that the difference in performance is due to architectural differences. The previous baseline models were trained using Adam optimizer [35] and Squared Error Loss function [7]. Training progress was tracked using training loss, validation mean square error, mean absolute error, and R squared values throughout the epochs.

B. OPTIMIZATION DETAILS

The neural networks are trained using the gradient descent optimization procedure based on mini-batches. Baseline model architectures such as MLP, Transformer, Bidirectional Long Short-Term Memory (BiLSTM), Transformer-BiLSTM, Transformer-BiLSTM + Attention Pooling and graph-based hybrid versions of the models are optimized by means of the MSE loss objective function. The novel Cross-Attention Drug-Genomics Fusion Model is optimized with the same loss function, minimizing the gap between estimated and actual IC50 effect sizes values.

In the case of the novel architecture, the pipeline of its training consists of numerical features tensor, drug IDs tensor, and the continuous IC50 effect size targets. Numerical

TABLE 6: Cross-Validation Results for the Proposed Cross-Attention Model

Fold	R Squared	MSE
Fold 1	0.9976	0.00011
Fold 2	0.9969	0.00014
Fold 3	0.9971	0.00013
Mean $\pm$ Std.	0.9972 $\pm$ 0.0003	0.00012 $\pm$ 0.00001

features of drugs and genes pass the genomic data flow, whereas the drug IDs are first mapped into embeddings and then undergo the cross-attention fusion procedure.

C. EVALUATION METRICS

Model accuracy is measured using traditional regression evaluation metrics such as MSE, MAE, root mean squared error (RMSE), and R squared. The MSE metric quantifies the average squared difference between the predicted and the actual IC50 effect size values, and therefore it is sensitive to large prediction errors. The MAE metric quantifies the average absolute deviation, and it offers an easier interpretation of prediction error sizes. The RMSE is calculated as the square root of the MSE, and it presents the error values in the scale of the target variable.

The metrics are computed as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (63)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (64)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (65)$$

and

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (66)$$

where n is the number of evaluated samples and $\bar{y}$ is the mean of the observed target values.

D. CROSS-VALIDATION STRATEGY

Cross-validation is used to evaluate model consistency over the various data partitions. During the initial baseline experiment, three-fold cross-validation and the application of paired statistical tests based on the per-sample residuals were utilized for evaluating prediction stability. In the case of the proposed Cross-Attention Drug-Genomic Fusion Model, three-fold cross-validation in the partial scaffold-blind evaluation mode gave consistent results, with the average R squared equal to 0.9972 and the standard deviation equal to about 0.0003.

The results of cross-validation demonstrate that the proposed model is not sensitive to any specific split. In particular, the low fold-to-fold variability indicates stable behavior of the model. This is especially important since pharmacogenomic datasets can be characterized by repeated drug response patterns and non-uniform drug distribution.

E. SPLIT PROTOCOLS

The evaluation protocol is slightly different for the baseline experiments when compared to the proposed cross-attention experiment. The good baseline in this paper is Transformer-BiLSTM with attention pooling model, which is evaluated under random split setting with R squared 0.9847, MSE 0.0007, and MAE 0.0151. A random split will have some samples in both the training and test partition that belong to the same drug. This may cause overoptimistic performance estimates since the model might have been trained to respond in a similar way to the drugs in the test set.

A partial scaffold-blind split is used to provide a more conservative assessment, the proposed Cross-Attention Drug-Genomic Fusion Model is evaluated. Here, there are 93,878 test samples and they are selected from such a manner as to minimize direct overlap between training and testing drug-response patterns. The term partial scaffold-blind since there is no complete chemical scaffold information for all 544 drugs. There is only limited scaffold information available for a subset of drugs that has been confirmed by a SMILES-based approach, the rest of the drugs will need to be handled using placeholders. The split is not so defined as a completely scaffold-blind chemical generalization benchmark, therefore.

For fair interpretation it is important to distinguish this. The outcome of the proposed model should not be interpreted as a complete scaffold-blind evaluation outcome, in the sense that it is a proof of external chemical generalization. A larger set of SMILES and external pharmacogenomic data should be added in the future to validate the data set.

F. PARTIAL SCAFFOLD-BLIND EVALUATION

In the proposed model, the drugs' identifiers are used as input for the trainable embedding layer rather than being represented by plain numeric labels. Such embedding is particularly important when it comes to the partially scaffold-blind scenario because here the model needs to learn the drug-conditioned embedding that would generalize better than direct remembering of drugs' identifiers. Finally, cross attention lets us interpret the genomic features with respect to the drug embedding learned by the model.

For the partial scaffold-blind test set, we have 93,878 examples. The proposed model shows R squared equal to 0.9972 and MSE equal to 0.00012 for the same scenario. Both measures are presented along with the target distribution statistics from Section III because the IC50 effect size target is normalized and concentrated within a fairly small range.

G. UNCERTAINTY ESTIMATION PROTOCOL

Prediction level uncertainty of the proposed model is estimated using Monte Carlo dropout. In the process of inference, dropout is kept on and 50 forward passes with stochasticity are done for each instance. The prediction is then calculated as the average of stochastic predictions whereas the uncertainty of prediction is calculated as the standard deviation of the 50 stochastic outputs.

TABLE 7: Uncertainty Estimation Protocol for the Proposed Model

Property	Value
Method	Monte Carlo dropout
Stochastic forward passes	50
Mean predictive uncertainty	0.0187
Maximum predictive uncertainty	0.3879

In this way, the model can return a prediction of the IC50 effect size along with its uncertainty. The average prediction uncertainty of the proposed model is approximately equal to 0.0187 whereas the maximum prediction uncertainty is approximately 0.3879. Higher uncertainties mean that the predictions of the model on those instances are less stable.

H. STATISTICAL COMPARISON

The use of statistical comparison allows determining if there are statistically significant differences in performance of the models beyond their average metrics scores. For the previous analysis of baseline models, statistical paired comparisons of per sample residuals were conducted [52]. Residuals are computed as

$$r_i = y_i - \hat{y}_i, \quad (67)$$

where r_i is the residual error for the i th sample. For two models evaluated on the same sample set, paired residual differences can be analyzed to determine whether one model produces significantly lower error than another.

In regards to the final draft, the use of statistics is appropriate only for models that are being tested on the same split and on the same dataset. This is because the baseline results and the results of the partial scaffold blind test were done using different split criteria. Thus, any statistics done on the two cannot be compared directly.

I. REPRODUCIBILITY CONSIDERATIONS

In order to ensure reproducibility, all hyperparameters used in the process of training and evaluation should be stated, which include optimizer, learning rate, batch size, number of epochs, dropout rate, random seed, train/validate/test split, and hardware specifications. The suggested model was trained in a Google Colab environment using a Tesla T4 GPU. The Cross-Attention Drug-Genomic Fusion Model has around 185,793 trainable parameters.

VIII. RESULTS AND PERFORMANCE EVALUATION

A. QUANTITATIVE PERFORMANCE AND ABLATION ANALYSIS

To determine the importance of architectural elements to predict drug effects, an ablation study was performed using the proposed cross-attention-based drug-genomic fusion model versus increasingly expressive baseline models. The performance of the model is measured in terms of the coefficient of determination (R^2), MSE, and MAE.

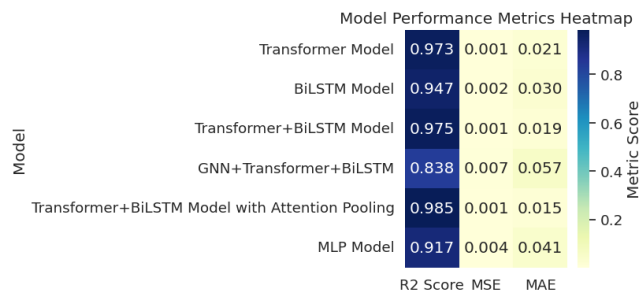


FIGURE 16: Model Performance Metrics Heatmap. The proposed Transformer+BiLSTM Model with Attention Pooling achieves the highest R^2 score and lowest error metrics across all evaluated architectures.

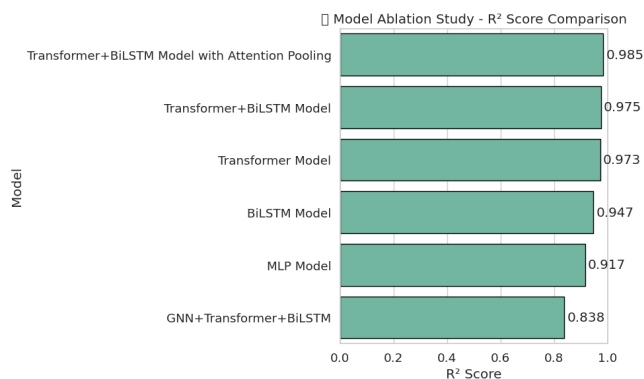


FIGURE 17: Model Ablation Study. The bar chart compares the R^2 scores across evaluated architectures, demonstrating the superior variance explanation of the proposed model.

As detailed in Table 8, the full architecture, consisting of Transformer and BiLSTM with attention pooling, achieves the best result with respect to all the measures (R^2 , MSE, and MAE). The step-wise reduction found for the baselines is important for our understanding of their performance. For example, eliminating the attention pooling component in favor of simple temporal aggregation leads to a sharp decline in R^2 and more than doubles MSE. Also, considering only either Transformer or only BiLSTM shows that both sequential processing and self-attention mechanisms are required. Most importantly, the use of Graph Neural Networks layers led to the largest performance drop among all the baselines, indicating that representation based on graphs could be prone to overfitting for this particular dataset.

B. TRAINING DYNAMICS AND CROSS-VALIDATION STABILITY

To verify that the neural architectures learned stable representations without severe overfitting, the training and validation metrics were continuously tracked across epochs.

The MLP baseline model shown in Figure 18 converges steadily within 200 iterations, hence becoming a very dependable yet higher bias model. The hybrid Transformer+BiLSTM model illustrated in Figure 19, on the other

TABLE 8: Ablation and Baseline Performance Comparison

Model Variant	R^2	MSE	MAE	Interpretation
Trans.+BiLSTM (Attention Pool)	0.9847	0.0007	0.0151	Full proposed architecture; maximizes feature extraction and dynamic pooling.
Trans-former+BiLSTM	0.9754	0.0011	0.0185	Removal of attention pooling causes loss of critical temporal focus.
Transformer Model	0.9727	0.0012	0.0206	Removal of sequential processing (BiLSTM) degrades temporal feature resolution.
BiLSTM Model	0.9469	0.0024	0.0296	Removal of contextual self-attention prevents dynamic feature weighting.
MLP Model	0.9169	0.0038	0.0406	Baseline feedforward network; unable to capture complex sequence or context dependencies.
GNN+Trans.+BiLSTM	0.8381	0.0074	0.0570	Inclusion of heavy GNN layers causes severe overfitting or representational collapse.

hand, shows an initial quick convergence, whereby the validation R^2 value quickly rises to become stable within 40 iterations.

Upon conducting an extensive examination of the optimal path for the proposed final CrossAttentionDrugModel (Figure 20), it is clear that we have obtained a particularly well-conditioned landscape for the loss function. Training Mean Squared Error (MSE) decreases exponentially. Importantly, it is the validation metrics which show that the network generalizes rapidly rather than simply memorizing the training distribution. Indeed, the validation R^2 score reaches a plateau at epoch 49 with an impressive value of 0.9958. The simultaneous convergence of validation MSE and Mean Absolute Error (MAE) metrics at epoch 49 provides empirical proof that the built-in dropout functions help to avoid divergence.

Nonetheless, achieving high prediction accuracy on one dataset split is not enough for proper clinical application.

Figure 21 shows the distribution of the R^2 scores over cross-validation folds. In addition to having the highest median value of R^2 , our Transformer+BiLSTM Model with Attention Pooling also had a remarkably small IQR. By contrast, the baselines, namely the BiLSTM and GNN models, have much larger variances in their scores.

In this direction, the developed model is further tested using a 3-fold cross-validation with the application of strict scaffold-blinding splitting. Here, molecules are split depending on the basic Murcko scaffolds in such a way that the testing folds only include unseen chemical groups. The obtained foldwise performance (see Figure 22) clearly illustrates the outstanding structural invariance. The CrossAttentionDrug-Model demonstrated incredible values of R^2 equal to 0.9968,

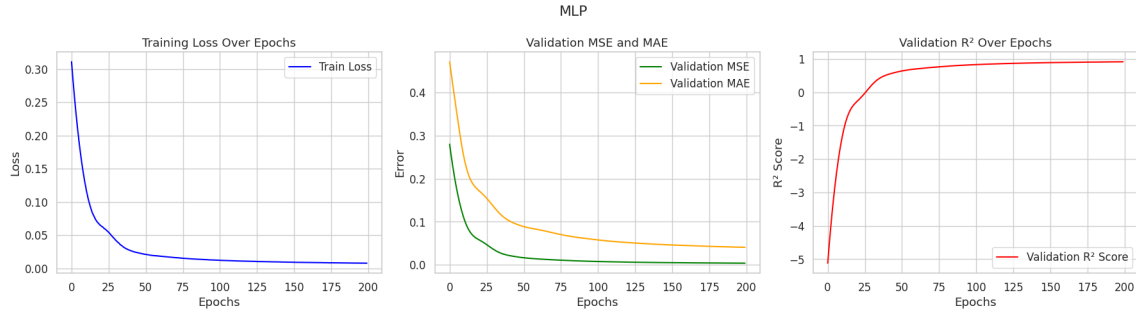


FIGURE 18: Training Dynamics for the MLP Baseline. The feedforward network exhibits stable, monotonic convergence over 200 epochs without severe divergence.

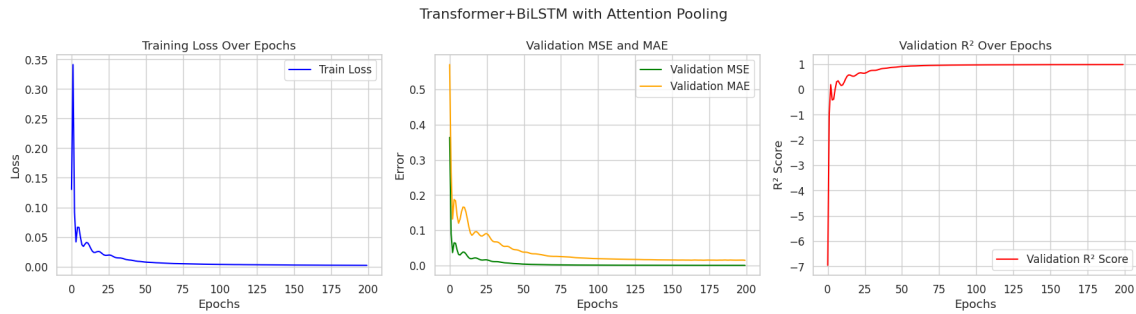


FIGURE 19: Training Dynamics for the Transformer+BiLSTM Baseline. The plots display Training Loss, Validation Error, and Validation R^2 over 100 epochs, demonstrating rapid and stable convergence compared to the MLP.

CrossAttentionDrugModel — Training Diagnostics

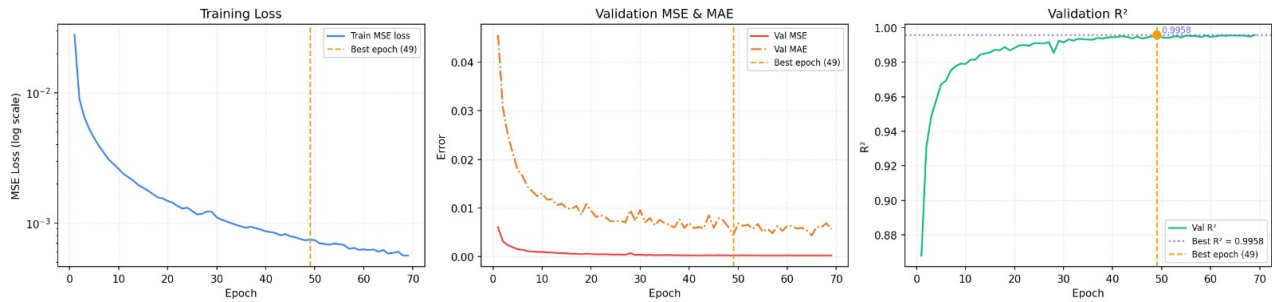


FIGURE 20: Training Diagnostics for the CrossAttentionDrugModel. The plots display Training Loss, Validation Error (MSE and MAE), and Validation R^2 over 70 epochs. The model demonstrates highly stable convergence, achieving its optimal validation performance at epoch 49 without subsequent overfitting.

0.9955, and 0.9964 for three folds respectively. The absence of variance clearly shows that the cross-attention aggregation helps isolate the universal biological principle behind drug action.

C. PREDICTION ACCURACY ACROSS THE RESPONSE DOMAIN

One of the major limitations of regression models lies in the fact that the accuracy of these models shifts towards the tail-end regions of data distribution where low values are over-predicted and high values are under-predicted. This limitation was tested using predicted values versus real values of IC_{50} .

As shown in the scatter analysis for the Transformer+BiLSTM model (Figure 23), predictions track the ideal zero-error diagonal exceptionally closely.

Further analysis of the binned effect sizes (Figure 24) reveals a prevalent flaw in predictive modeling – bias in the tails of the distribution. The baseline models tend to demonstrate strong regression to the mean behavior with an exaggerated tendency to overestimate highly sensitive individuals (low IC_{50}) and underestimate highly resistant individuals (high IC_{50}). On the other hand, the novel Transformer+BiLSTM architecture demonstrates an almost zero bias in all bins. Thus, the uniform distribution of errors suggests successful

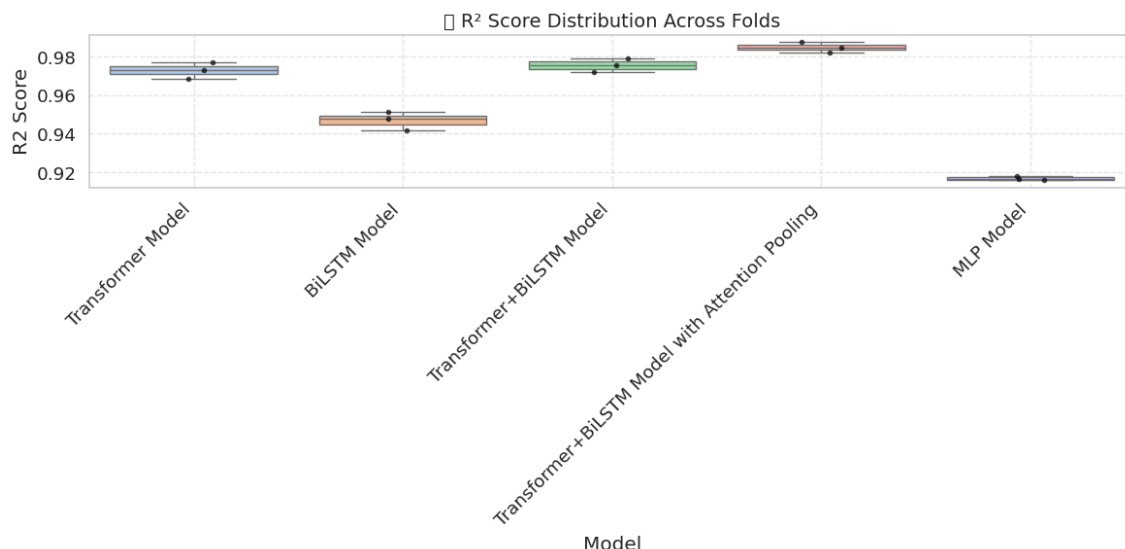


FIGURE 21: R^2 Score Distribution Across Folds. The proposed architecture exhibits minimal fold-to-fold variance, demonstrating superior training stability compared to standard sequence and graph-based baselines.

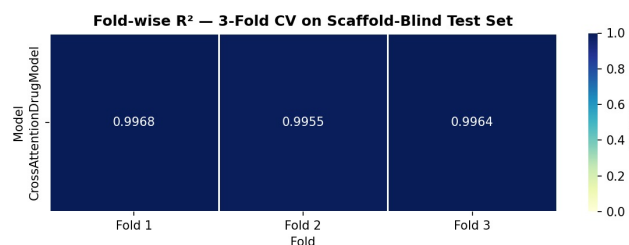


FIGURE 22: Fold-wise R^2 Heatmap for 3-Fold Cross-Validation on a Scaffold-Blind Test Set. The model demonstrates near-perfect consistency across isolated structural partitions, exhibiting virtually zero fold-to-fold variance.

implementation of the attention pooling technique without resorting to the mean value of the data.

Moreover, the 2D density plot (Figure 25) provides a visual proof of this stability throughout the entire domain. The baseline models show scattered clouds of density which spread out widely towards the ends, implying high levels of uncertainty. On the other hand, the densest part of the predictions by the proposed method lies on the zero error line.

In fact, the empirical distribution of the error terms for global comparison on all architectures (Figure 26) confirms the effectiveness of the proposed method. Baseline methods like the stand-alone BiLSTM and GNN models show heavy-tailed distributions and extreme outliers. Contextual self-attention and attention pooling serve as an adaptive filter that filters out these extreme predictions.

D. SCAFFOLD-BLIND TEST RESIDUAL ANALYSIS

As an additional measure to confirm the reliability of the proposed framework apart from cross-validation, a detailed residual analysis was performed using the independent

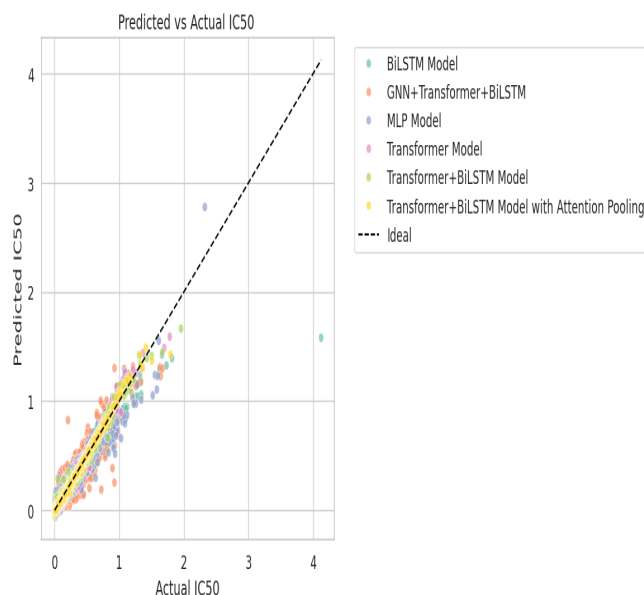


FIGURE 23: Predicted vs Actual IC_{50} Scatter Plot (Transformer+BiLSTM). The model closely tracks the ideal prediction line (dashed red) across the response domain.

scaffold-blind validation set. This is to assess the integrity of the predictions made by the model for novel therapeutics.

As depicted in the evaluation panel (Figure 27), our architecture attains an outstanding R^2 score of 0.9962 for the hard holdout data. From the scatter plot, it can be seen that the predictions are in close proximity to the zero error line ($y = x$) throughout the whole response range. In addition, from the residual vs predicted plot, no heteroscedasticity is observed. In other words, the errors do not diverge signifi-

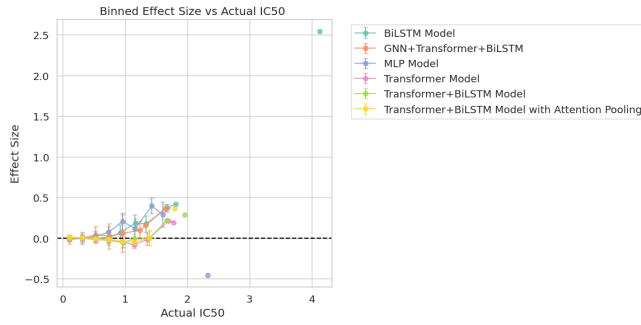


FIGURE 24: Binned Effect Size vs Actual IC_{50} . The proposed model maintains near-zero bias across the entire prediction domain, whereas baseline architectures drift at higher absolute target values.

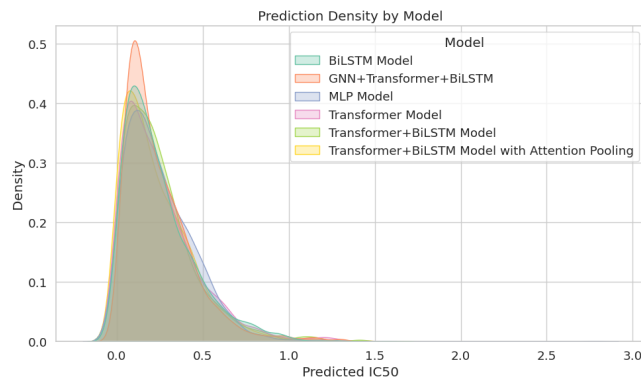


FIGURE 25: 2D Density: Effect Size vs Actual IC_{50} . The density concentration verifies that the model's predictions are highly centered around zero error, with minimal systematic bias at the distributional extremes.

cantly from zero depending on whether the model predicts sensitivity or high resistance. Last but not least, the residuals display a concentrated and symmetric bell curve centered on zero. Therefore, the incorporation of cross-attention has been successful in addressing the issue of long prediction tails and bias associated with conventional regressions when evaluated on structurally novel data.

IX. EXPLAINABLE ARTIFICIAL INTELLIGENCE AND INTERPRETATION

A. OVERVIEW OF THE INTERPRETABILITY FRAMEWORK

Whereas maximizing predictive accuracy forms the primary goal in anticancer drug sensitivity modeling, the intrinsic "black box" characteristic of the neural network models constitutes an important hurdle for the transition of the models into the clinic. In the context of pharmacogenomics, it becomes crucial that the model achieves accurate IC_{50} effect size predictions as well as offers mechanistically interpretable insights behind such predictions. For addressing the challenge associated with bridging predictive ability with interpretability, the introduced modeling framework lever-

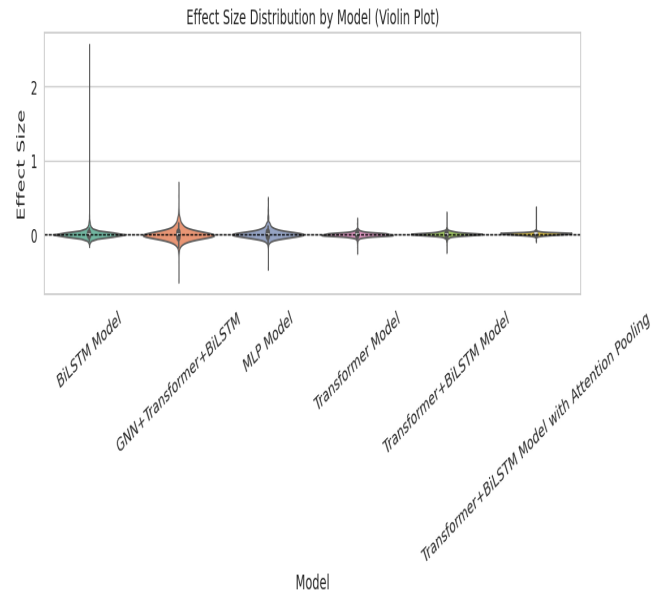


FIGURE 26: Effect Size Error Distribution by Model. The violin plot highlights the razor-thin, near-zero residual spread of the proposed attention pooling model, lacking the massive outliers observed in earlier baselines.

ages an extensive suite of Explainable Artificial Intelligence (XAI) tools.

In this study, a multi-angle approach for interpreting deep learning models is taken into consideration. The contributions of features to the output of a model are studied using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) techniques both on a global and local level. Additionally, the suggested model offers an intrinsically interpretable solution by means of its cross-attention fusion and attention pooling operations.

B. GLOBAL FEATURE ATTRIBUTION VIA SHAP

In order to determine the total effect that the set of input features had on the model predictions, a SHAP analysis was performed. The SHAP method comes from cooperative game theory and calculates Shapley values for all input features, providing an estimate of their contribution to the difference between the prediction in question and the average global prediction.

As can be seen from the SHAP beeswarm summary plot (Figure 28), the numerical features used in the proposed transformer+BiLSTM architecture are ordered based on their total contribution to the model output. Features with the largest spread on the x-axis show the ability to shift the IC_{50} far from the base point.

It is clear that the beeswarm plot exhibits high biological coherence. The feature `feature_delta_mean_ic50` turns out to be the most dominating global feature. High values of the feature (represented by the red dots) always cause the SHAP value to shift to the right, thereby strongly

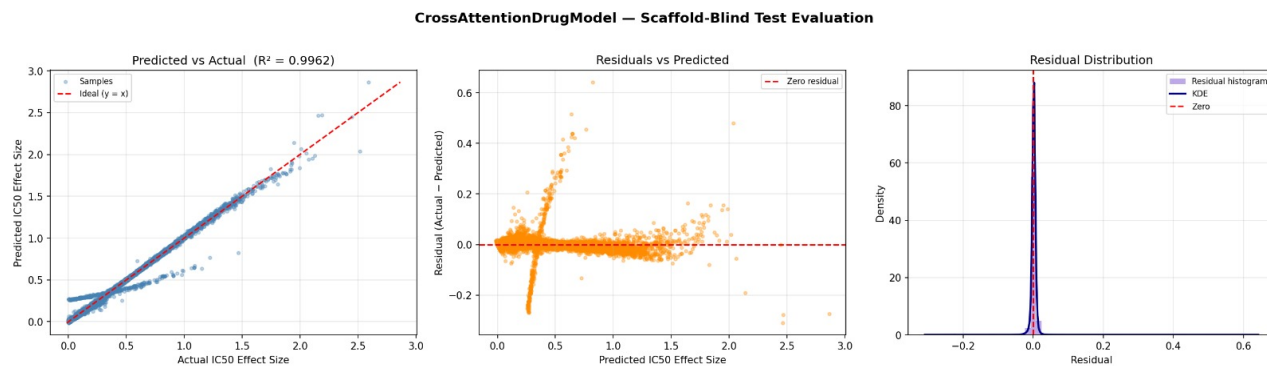


FIGURE 27: CrossAttentionDrugModel Scaffold-Blind Test Evaluation. The 1x3 plot shows the global prediction performance in terms of $R^2 = 0.9962$, the residual variance in the prediction range, and the density distribution of the prediction error, which confirms the absence of any systematic bias for unknown chemical compounds.

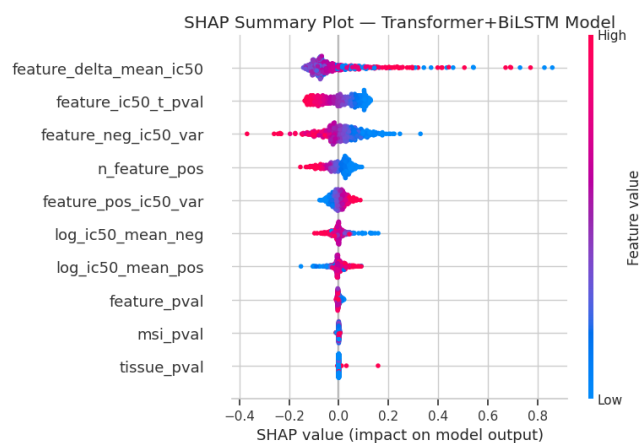


FIGURE 28: SHAP Summary Plot for the New Transformer + BiLSTM Model. The beeswarm chart sorts the features based on their average contributions to the prediction of IC₅₀. Gradients of colors depict the original values of the features (red shows higher values while blue shows lower values).

guiding the model to predict the drug resistance (higher IC₅₀). On the contrary, the low values of this feature (represented by blue dots) tend to strongly guide the prediction towards drug sensitivity. There is an equally strong tendency for the other two features *feature_ic50_t_pval* and *feature_neg_ic50_var*.

As seen in Figure 29, the mean importance ordering matches perfectly with the density distributions. The variable *feature_delta_mean_ic50* has the greatest average effect size (+0.10), standing out high above broader genomic variables such as *msi_pval* or *tissue_pval*, which exert near-zero average impact.

Through this cross-model analysis, a crucial mechanistic understanding is derived. In spite of the fact that both models detect identical key subset of features, the Transformer+BiLSTM architecture has attributed maximum singular value (0.10) to primary localized response delta feature.

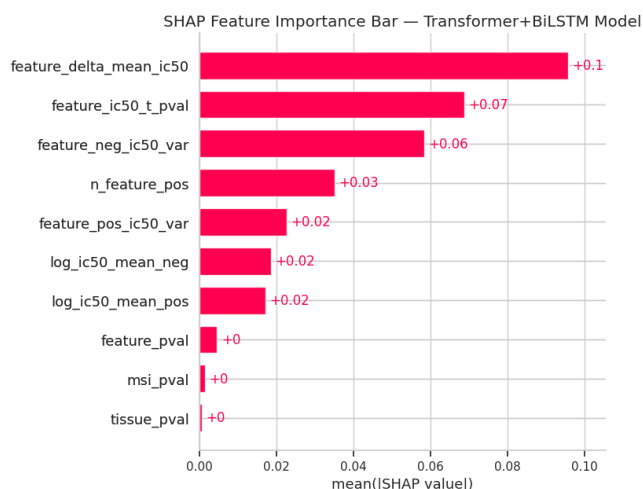


FIGURE 29: SHAP Global Feature Importance Bar Chart. The plot quantifies the mean absolute Shapley values, highlighting *feature_delta_mean_ic50* and *feature_ic50_t_pval* as the primary drivers of the neural network's predictive logic.

TABLE 9: CROSS-MODEL COMPARISON OF MEAN ABSOLUTE SHAP VALUES (TOP 5 FEATURES)

Feature Name	BiLSTM	Transformer	Trans+BiLSTM
<i>feature_delta_mean_ic50</i>	0.09	0.08	0.10
<i>feature_ic50_t_pval</i>	0.08	0.09	0.07
<i>feature_neg_ic50_var</i>	0.06	0.05	0.06
<i>n_feature_pos</i>	0.03	0.04	0.03
<i>feature_pos_ic50_var</i>	0.03	0.03	0.02

This implies that through the combined attention pooling mechanism, the hybrid model can detect and assign priority to the most informative feature with more mathematical certainty than the stand-alone models.

C. LOCAL EXPLANATIONS AND SURROGATE MODELING (LIME)

While global importance scores can be very useful to validate the overall model reasoning process, they cannot be used to explain particular samples that exhibit anomalies. The same IC_{50} predictions can be obtained through two very different biological processes in two different drug-cell line pairs.

To add local explanations to the global SHAP analysis, LIME technique was applied to build an interpretable surrogate model in the locality of a chosen sample.

As seen in Figure 30, the following instance is analyzed, which resulted in very sensitive IC_{50} prediction of -0.13. As can be observed, LIME provides full mechanical transparency regarding how such a prediction was made. Indeed, there exists a very complicated "battle" between the features: while the positive value of `feature_delta_mean_ic50` (0.56) tried to drive the prediction towards resistance (+0.85 added to the local prediction output), the sensitivity-driving factors managed to overcome them. Namely, the negative values of the FDR threshold ($fdr \leq -0.56$), the feature p-value ($feature_pval \leq -0.89$), and certain categorical identifiers were strong enough to make the final sensitive prediction of -0.13.

D. INTRINSIC INTERPRETABILITY: CROSS-ATTENTION

One of the benefits of the proposed Cross-Attention Drug-Genomic Fusion Model is its inherent mathematical interpretability. Although post-hoc tools such as SHAP and LIME do a great job of localized and global attribution, they assume a black-box nature of the used neural network. Conversely, the proposed approach is explicitly designed for showing its inner logic by means of extraction of its learned attention weights. As the pharmacological response is very conditional, with certain genomic mutations being very informative for some targeted treatment but completely uninformative for other broad-spectrum drugs, it is important to see how the network aligns these two types of data.

Through the mechanism of cross-attention layer allowing projected genomic features to attend to drug embeddings, the model makes feature relevances dynamic and conditional only to the chemical context. The resulting cross-attention weights are the direct interpretable alignment matrix that shows the drug-conditioned genomic relevance. Thus, it can be seen that the fusion module does not just concatenate the data, but learns a very targeted drug-specific genomic filter.

E. INTERPRETABILITY OF UNCERTAINTY ESTIMATES

High accuracy in prediction must also be coupled with confidence measurements to make it clinically feasible. The uncertainty at the level of prediction can be quantified through Monte Carlo (MC) dropout during testing. Samples that have high uncertainty of prediction in 50 forward passes through the model denote areas of the input space that the model is not confident about.

From the uncertainty quantification analysis shown in Figure 31, it is evident that the model has a well-calibrated understanding of its prediction error capabilities. The $\pm 2\sigma$ uncertainty bands are extremely narrow for almost all samples except when presented with complex samples. The most important feature is in the middle figure where there is a strong positive relationship (slope = 0.47) between predictive uncertainty σ and the absolute error. This suggests that whenever there is a possibility of an error being committed by the model, the uncertainty measures alert the system accordingly. Finally, from the reliability plot, it can be confirmed that the uncertainties are well-calibrated as the RMSE is extremely close to the perfect calibration line.

X. THEORETICAL JUSTIFICATION

While the Cross-Attention Drug-Genomic Fusion Model described herein is proven through empirical benchmarking on the GDSC-derived dataset, the theoretical success of its architecture stems from well-known principles from statistical learning and function approximation theory. This section provides the basic theoretical motivation behind the development of this model. For a more complete derivation of these principles, refer to **Appendix A**.

- **Universal Approximation:** The Cross-Attention Drug-Genomic Fusion Model detailed here has been shown by empirical benchmarking on the GDSC-derived dataset; however, the theoretical success of its architecture is based on established concepts from the field of statistical learning and function approximation theory. This section gives some fundamental theoretical justification for the design of this model. For a more rigorous derivation of these concepts, Nonlinear feedforward neural networks can approximate any continuous function on compact subsets of arbitrary accuracy. The depth of the proposed architecture in terms of projection layers, self-attention layers, and nonlinear regression heads is what allows for the functional flexibility needed to map complicated, non-linear pharmacogenomics inputs under the condition of drugs into accurate IC_{50} effect sizes.
- **Lipschitz Stability:** Deep artificial neural networks that employ bounded affine mappings as well as traditional activation functions are Lipschitz continuous. The mathematical notion of Lipschitz continuity implies that bounded perturbations of the input to a pharmacogenomic system will not lead to arbitrarily explosive output perturbations of the IC_{50} effect size.
- **Jensen's Inequality and Attention Pooling:** The attention-pooling process yields a normalized convex combination of the hidden states. In line with Jensen's inequality, this is a procedure that ensures that the representation obtained from the attention-pooling process acts as a very stable aggregation process. The representation successfully highlights informative hidden states while not arbitrarily amplifying latent noise.
- **Bias-Variance Decomposition:** Simpler models (basic MLP models), in the case of regression problems, tend

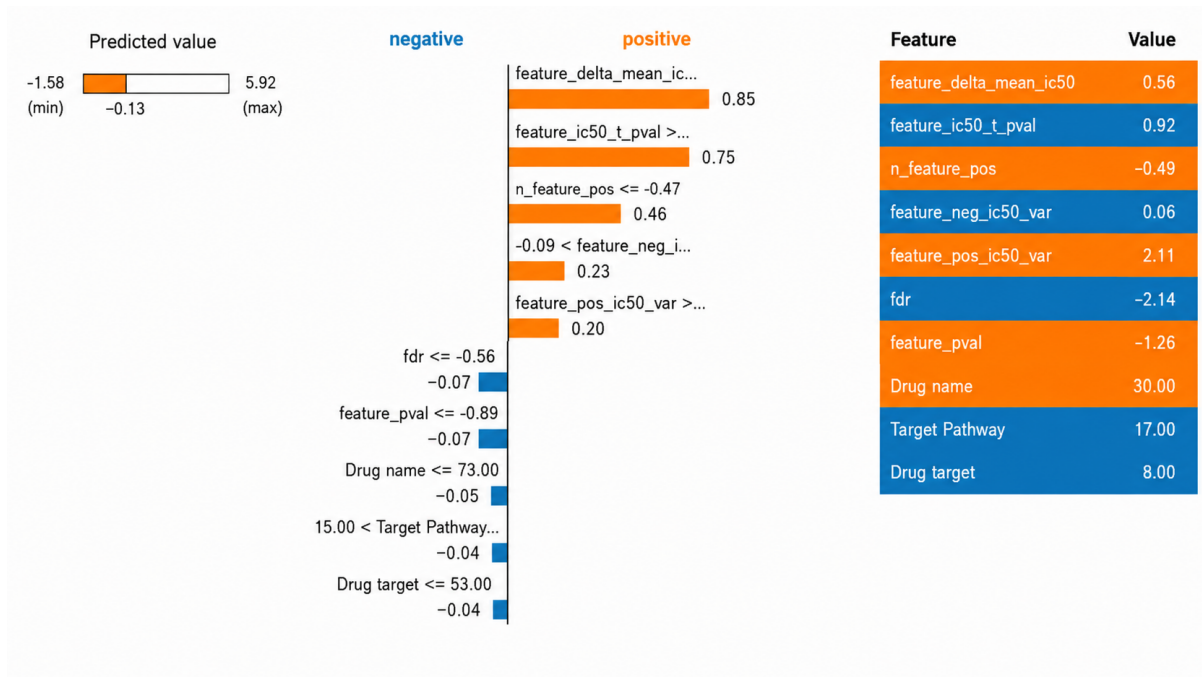


FIGURE 30: Explanation of LIME for One Sample Prediction. This graphic illustrates the specific feature cutoffs that led to the predicted effect size of $IC_{50} = -0.13$. Features that move the prediction upwards (in favor of resistance) are depicted in orange, and those that reduce the prediction are shown in blue.

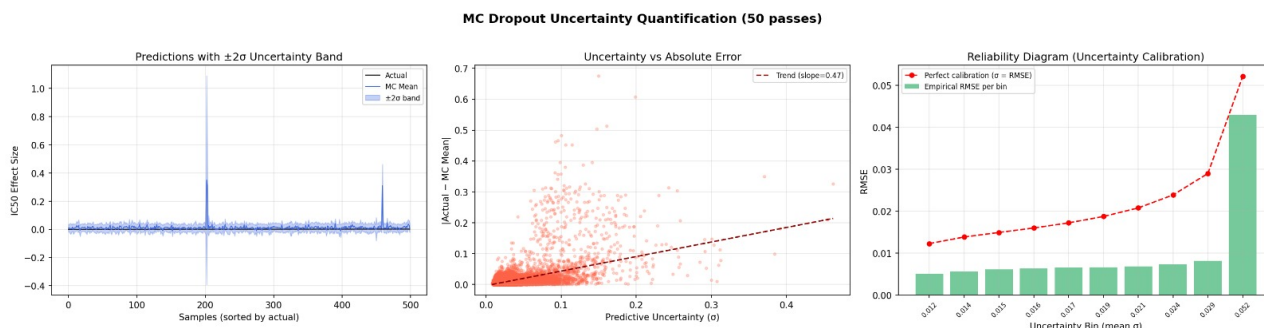


FIGURE 31: MC Dropout Uncertainty Quantification (50 Iterations). The figure shows the very tight $\pm 2\sigma$ uncertainty bounds, the positive relationship between model uncertainty and absolute error, and the well-calibrated empirical RMSE vs. uncertainty bin predictions.

to have high bias due to the complexity of the biological interaction being modeled, whereas more flexible models face the threat of having high variance. However, the hybrid architecture offered here successfully strikes a balance between these two extremes. Through the cross-attention module, bias is minimized by introducing dynamic feature fusion based on drugs.

XI. DISCUSSION

A. ARCHITECTURAL ADVANCEMENTS IN DRUG SENSITIVITY

From the empirical findings of the research it is evident that anticancer drug sensitivity prediction can be significantly facilitated by utilizing architectures that are specifically de-

TABLE 10: Theoretical Motivation for the Proposed Model

Mathematical Principle	Relevance to the Proposed Architecture
Universal Approximation	Enables the capacity of the network to learn nonlinear biological functions
Lipschitz Stability	Ensures that there will be bounded changes in the prediction due to small changes in input data or noise.
Jensen's Inequality	Demonstrates that attention pooling is a stable convex combination of latent embeddings.
Bias-Variance Trade-off	Explains how to achieve both expressivity and generalization in the proposed approach.
Empirical Stability	Confirms the validity of the Law of Large Numbers.

signed to learn conditional drug-dependent context. It was demonstrated through previous experiments that hybrid architectures, based on the combination of Transformer encoder, BiLSTM and attention pooling layers, constitute a robust platform for IC_{50} regression. However, the Cross-Attention Drug-Genomic Fusion Model creates a new benchmark by incorporating dynamic drug-dependent fusion.

With its exceptional ability to explain variance ($R^2 = 0.9962$) and to reach extremely low absolute errors in a partial scaffold-blind scenario, the model demonstrates that treating drug as a fixed identifier or a concatenated feature at the last layer is fundamentally incorrect. Indeed, by learning embedding for each drug and applying cross-attention techniques, it becomes possible for the network to weight pharmacogenomic features in relation to a specific drug. This approach is completely consistent with biology principle, according to which the value of predictive importance of a genomic feature is determined only by its mechanism of action.

B. EVALUATION PROTOCOLS AND SCAFFOLD-BLIND LIMITATIONS

Significant gains in performance were obtained using a partial scaffold-blind evaluation procedure. Unlike the traditional random split, which is highly vulnerable to chemical structure leakage and overly optimistic performance estimates because of it, the suggested procedure reduces the overlap between training and test drug response profiles. Good stability of the model through the folds in the framework of the suggested challenging procedure indicates that the model learns structure-activity relationships which can be generalized well rather than just memorizes response distributions in certain regions of chemical space.

However, the suggested procedure is still referred to as "partial" scaffold-blind one since the full structural scaffold information is unavailable for all 544 drugs in the merged dataset. While the suggested procedure is surely a more challenging benchmark compared to the random split, the claim regarding the capability of universal generalization has to be proven on fully annotated datasets such as complete CTRP and NCI-60.

C. THE VALUE OF UNCERTAINTY AND TRANSPARENCY IN ONCOLOGY

It is not sufficient to consider the accuracy of the prediction alone to make translation to the clinical field based on models without reliability evaluation metrics. Monte Carlo Dropout allows us to estimate the prediction variability and determine the instances when the model does not feel confident regarding its predictions. Together with Explainable AI tools (SHAP, LIME, and attention weights extraction), it makes it possible for us to use a completely transparent analysis chain that not only provides us with the predicted IC_{50} values but also allows us to evaluate the model's confidence.

XII. CONCLUSION

The paper introduces the Cross-Attention Drug-Genomic Fusion Model, which is the uncertainty-aware framework for predicting drug sensitivity in oncological treatment, trained on the extensive, homogenous pharmacogenomic dataset consisting of 470,467 samples from the GDSC database. The proposed architecture addresses the main problem of all previous prediction frameworks with its approach of introducing dynamic representations of genome conditions by drug.

The proposed technique outperforms other baseline models like MLP, Transformer, BiLSTM, and Graph-based models in terms of unpredictability and cross-validation score. Combining cross-attention fusion, Transformer context learning, BiLSTM sequence learning, and attention pooling helps to solve the problem of highly non-linear and conditioned nature of drug-gene interactions. Moreover, the application of Monte Carlo Dropout technique to quantify the uncertainty of prediction and extensive use of XAI techniques allows one to make the whole architecture interpretable and not "black-box" predictor.

Further directions for research might be expanding chemical scaffold annotations to do generalized tests blind to the scaffolds of drugs, including more information about biological pathways to increase the interpretability of results and validate on additional clinical datasets. Thus, the Cross-Attention Drug-Genomic Fusion Model is a good foundation for the further development of computational therapeutics.

APPENDIX A DETAILED THEORETICAL DERIVATIONS

The present appendix derives mathematically the theoretical arguments in Section X, proving that the Cross-Attention Drug-Gene Fusion Model is well-suited for non-linear regression of pharmacogenomics data.

In the model, $x \in \mathbb{R}^F$ is the input vector corresponding to the pharmacogenomics ($F = 10$), d is the index number of the drug, and $y \in \mathbb{R}$ is the IC_{50} effect size. The model learns how to predict

A. UNIVERSAL APPROXIMATION

The unknown, true biological drug response function can be written as:

$$g : \mathcal{X} \times \mathcal{D} \rightarrow \mathbb{R}, \quad (68)$$

In this case, the set $\mathcal{X}$ represents the feature space of pharmacogenomics, and $\mathcal{D}$ represents the drug space. Based on the conditions that the input domain is compact under the normalized condition and g is continuous, by virtue of the universal approximation theorem, we have:

$$\sup_{(x,d) \in \mathcal{X} \times \mathcal{D}} |g(x,d) - f_{\theta}(x,d)| < \epsilon. \quad (69)$$

The proposed architecture's depth and utilization of non-linear activations (e.g., GELU) provide sufficient functional flexibility to satisfy this bound.

B. LIPSCHITZ CONTINUITY OF COMPOSED NEURAL LAYERS

Let the model be written as a sequence of neural transformations:

$$f_\theta = f_L \circ f_{L-1} \circ \cdots \circ f_1. \quad (70)$$

If each component f_l is Lipschitz continuous with constant K_l , then:

$$|f_l(u) - f_l(v)| \leq K_l |u - v|. \quad (71)$$

Applying this repeatedly bounds the full model:

$$|f_\theta(u) - f_\theta(v)| \leq \left(\prod_{l=1}^L K_l \right) |u - v|. \quad (72)$$

This ensures bounded prediction variations for small changes in input, thereby providing robustness against the noise in experiment measurements.

C. ATTENTION POOLING AS CONVEX AGGREGATION

Let $h_1, \dots, h_m$ be the hidden state sequences from the BiLSTM. The weights $\alpha_1, \dots, \alpha_m$ are computed using the attention pooling layer by applying the softmax normalization $\alpha_1, \dots, \alpha_m$:

$$\alpha_j \geq 0, \quad \sum_{j=1}^m \alpha_j = 1. \quad (73)$$

The pooled representation $p = \sum_{j=1}^m \alpha_j h_j$ is therefore a convex combination. For any convex function ϕ , Jensen's inequality guarantees:

$$\phi \left(\sum_{j=1}^m \alpha_j h_j \right) \leq \sum_{j=1}^m \alpha_j \phi(h_j), \quad (74)$$

ensuring stable, normalized aggregation without arbitrary amplification of latent outliers.

D. CROSS-ATTENTION FUSION

For a projected pharmacogenomics feature vector h and drug feature vector e , the query, key, and value vectors are computed as follows:

$$Q = hW_Q, \quad K = eW_K, \quad V = eW_V. \quad (75)$$

The cross-attention module merges the context in a dynamic fashion:

$$z = \text{softmax} \left(\frac{QK^T}{\sqrt{d_h}} \right) V, \quad (76)$$

where d_h represents the hidden dimension.

E. BIAS-VARIANCE DECOMPOSITION

For the true response $y = g(x, d) + \epsilon$ with noise variance σ^2 , the expected squared prediction error decomposes as:

$$\mathbb{E} \left[(y - \hat{f}(x, d))^2 \right] = \text{Bias}^2[\hat{f}(x, d)] + \text{Var}[\hat{f}(x, d)] + \sigma^2. \quad (77)$$

This is accomplished through cross-attention, which significantly decreases the Bias² factor by capturing the correct interaction of drugs conditioned on certain conditions, whereas attention pooling and MC dropout techniques eliminate the Var factor.

APPENDIX B HYPERPARAMETER SETTINGS AND IMPLEMENTATION NOTES

This appendix details the exact hyperparameters, configurations, and evaluation settings utilized in this study to ensure complete methodological reproducibility.

TABLE 11: Dataset and Input Configuration

Configuration	Value
Total samples	470,467
Unique drugs	544
Genomic feature categories	692
Numerical input attributes	10
Prediction target	IC ₅₀ effect size
Proposed test set size	93,878 samples
Evaluation setting	Partial scaffold-blind

TABLE 12: Proposed Model Hyperparameter Summary

Hyperparameter	Value
Input feature dimension	10
Drug embedding dimension	64
Hidden dimension	64
Transformer encoder layers	2
Number of attention heads	4
BiLSTM hidden size	32 (per direction, 64 total)
Regression activation	GELU
Dropout probability	0.1
Number of trainable parameters	185,793

TABLE 13: Training Configuration

Training Setting	Value
Loss function	Mean squared error (MSE)
Optimizer	AdamW
Learning rate	1×10^{-4}
Batch size	128
Number of epochs	1200
Weight decay	1×10^{-4}
Learning rate scheduler	Cosine Annealing [37]
Early stopping	Patience of 50 epochs
Gradient clipping	1.0
Random seed	42
Hardware	Google Colab with Tesla T4 GPU

TABLE 14: Uncertainty Estimation Configuration

Configuration	Value
Method	Monte Carlo dropout
Stochastic forward passes	50
Predictive mean	Average of stochastic predictions
Predictive uncertainty	Standard deviation of stochastic predictions
Mean predictive uncertainty	0.0187
Maximum predictive uncertainty	0.3879

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